

Use of **population descriptors** and **gene-environment interactions** in **statistical genetics**

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Statistical Genetics Course 2026



Outline

1. Social factors in statistical genetics and genetic epidemiology

2. Use of population descriptors in human genetics

- What are population descriptors?
- Why and how do we use them?
- Practical considerations and real-life examples

3. Gene-environment interactions (GxE)

- What is GxE research? Why do we do it?
- State of the science
- Challenges of GxE studies
- Analysis & practicum

Outline

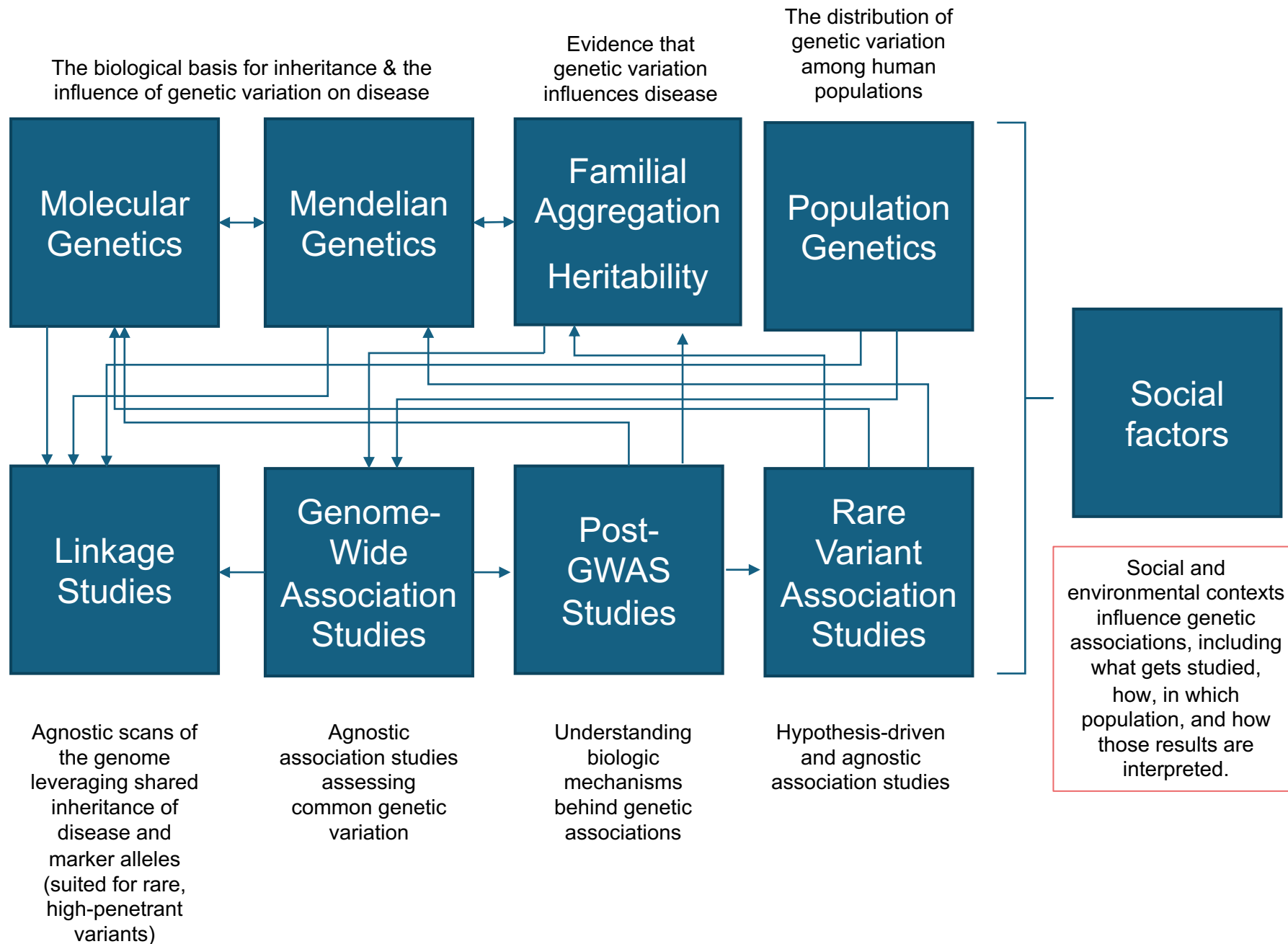
1. Social factors in statistical genetics and genetic epidemiology

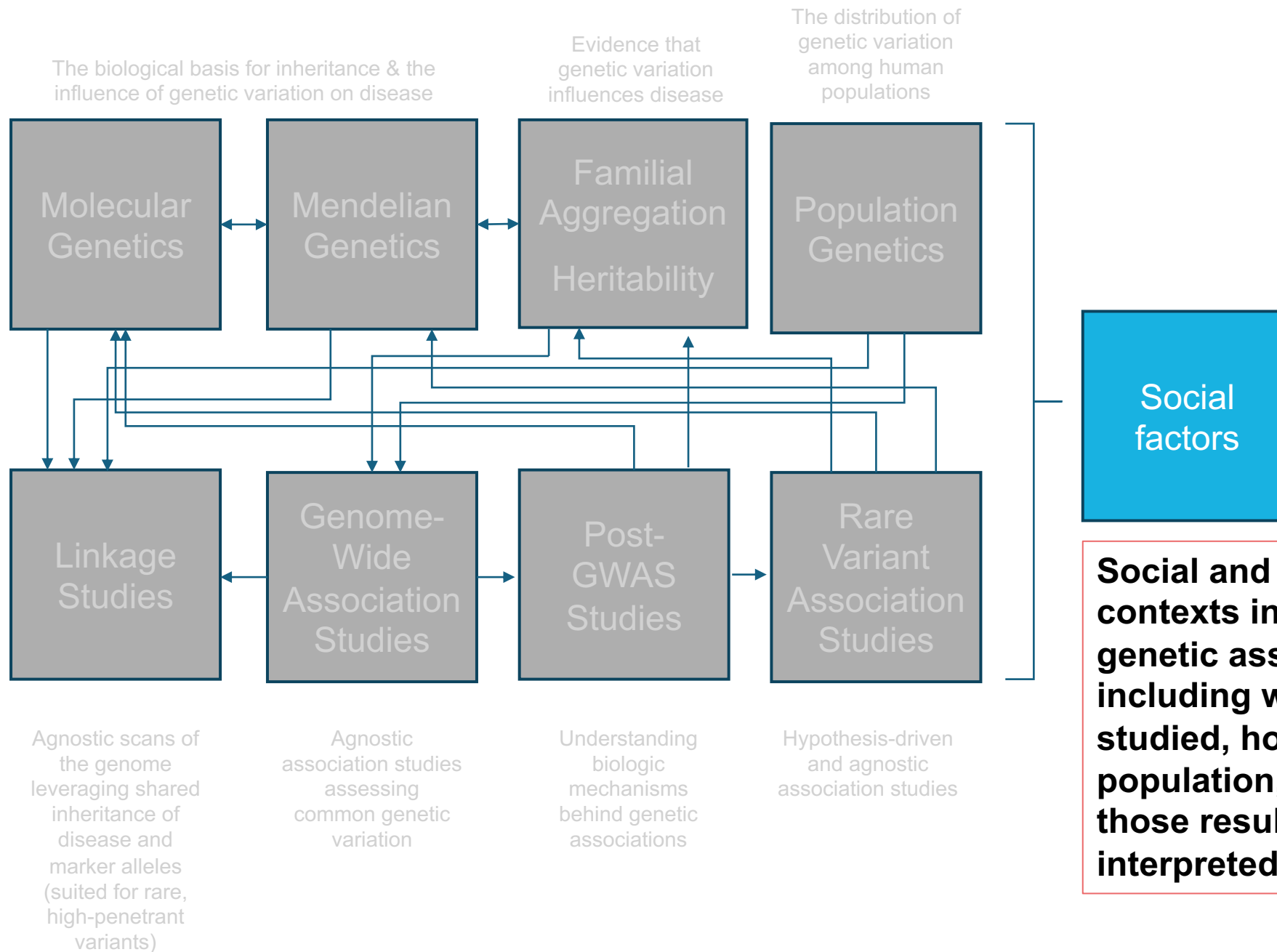
2. Use of population descriptors in human genetics

- What are population descriptors?
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3. Gene-environment interactions (GxE)

- What is GxE research? Why do we do it?
- State of the science
- Challenges of GxE studies
- Basic analysis framework & practicum





Population Descriptors

How do you define your population?

*Where are they geographically?
Are their identities self-reported, statistically assigned, etc.?*

**Social
factors**

Environmental Context

What do your population's lives look like?

*Where do they live? What environments are they exposed to?
What do they do in their daily lives?*

Social and environmental contexts influence genetic associations, including what gets studied, how, in which population, and how those results are interpreted.

Use of population descriptors

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What are population descriptors?

ways of describing or distinguishing people from each other based on perceived or actual differences

→ examples include race, ethnicity, nationality, and geographic location, although this is a non-exhaustive list

Definitions

Ancestry

a person's origin or descent, lineage, “roots,” or heritage, including kinship

Genetic ancestry

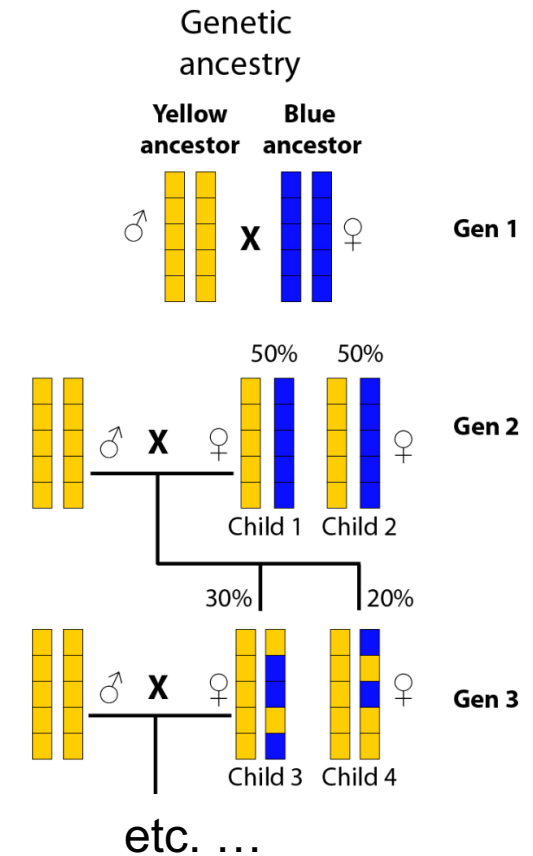
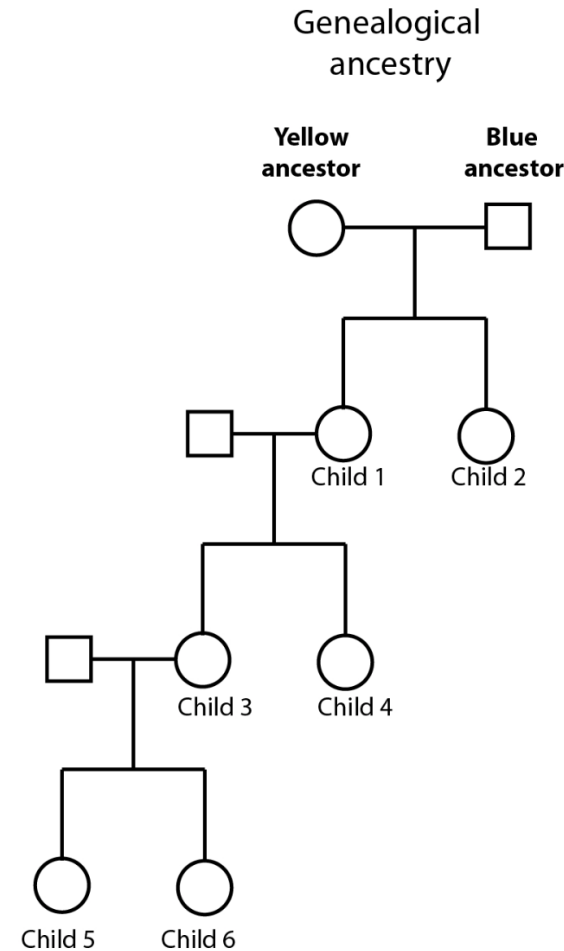
the paths through an individual's family tree by which they have inherited DNA from specific ancestors

Genetic similarity

quantitative measure of the genetic resemblance between individuals that reflects the extent of shared genetic ancestry

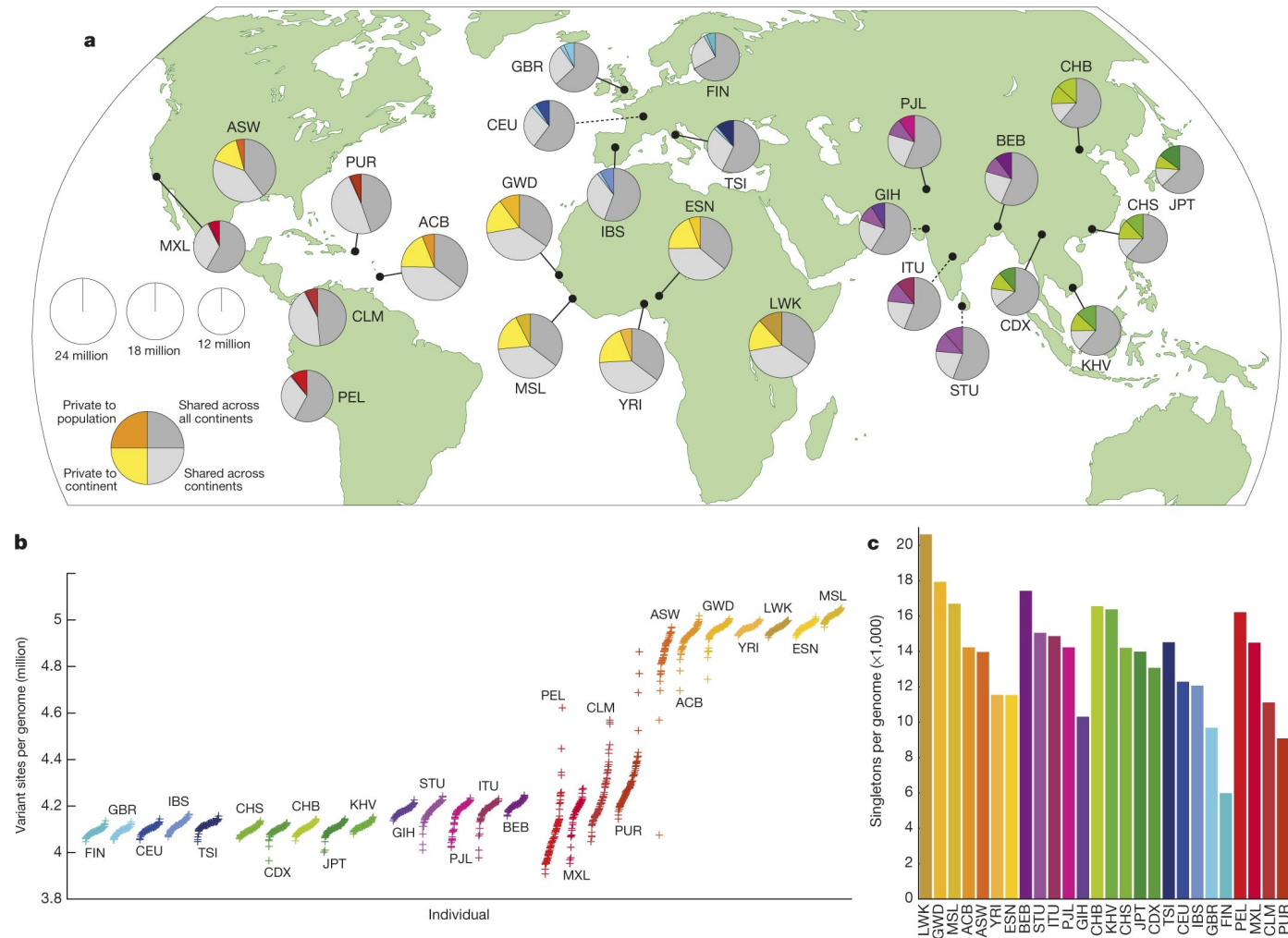
Genetic ancestry

- “stretches of the genome we inherit from our ancestors”
- (Mathieson “What is ancestry?” *PLOS Genet.* 2020)

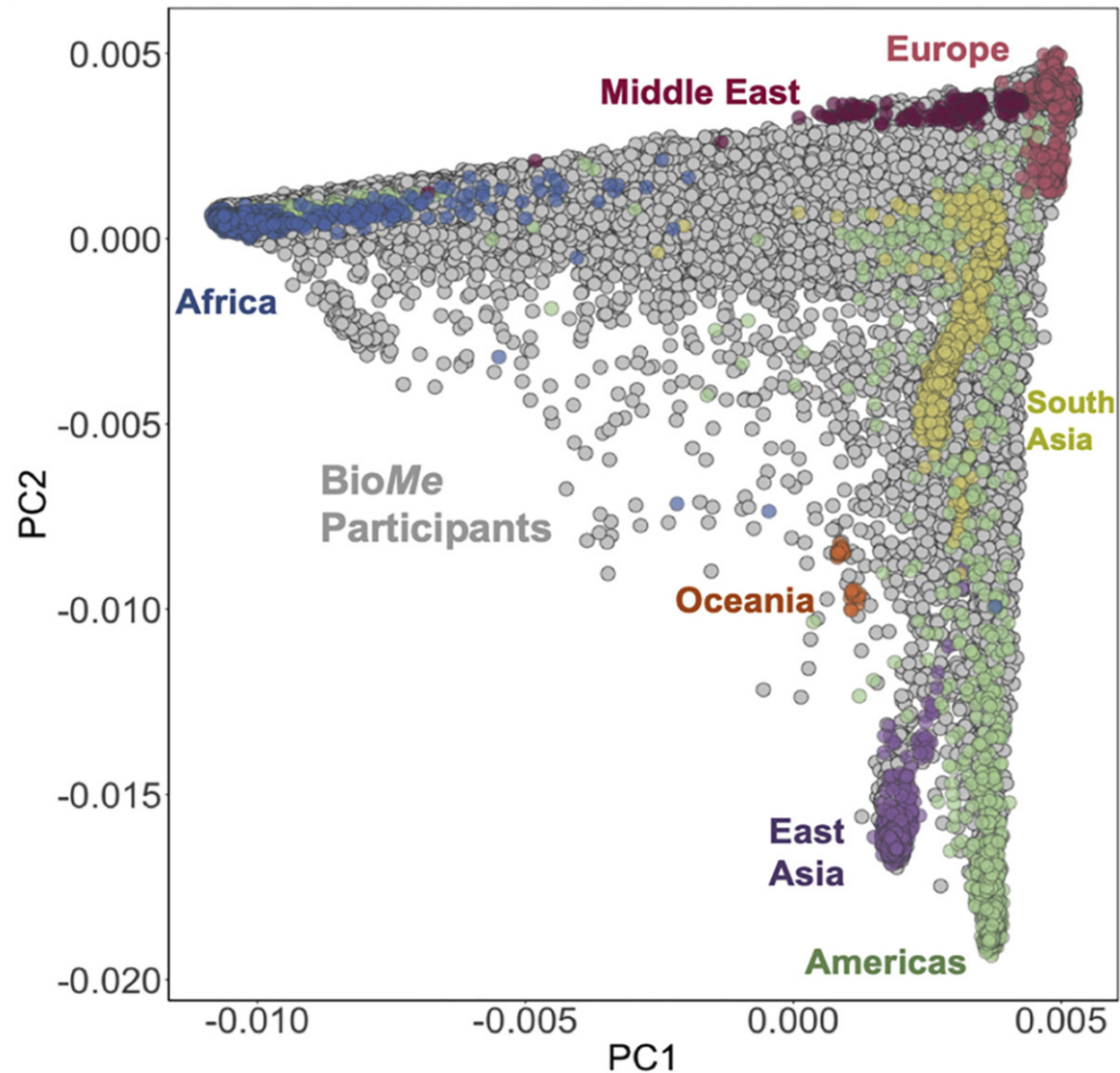


How is genetic ancestry operationalized?

Assigned through reference panels that tell us about the genetic background of people who are “representative” of an “ancestry group,” which have been arbitrarily designed



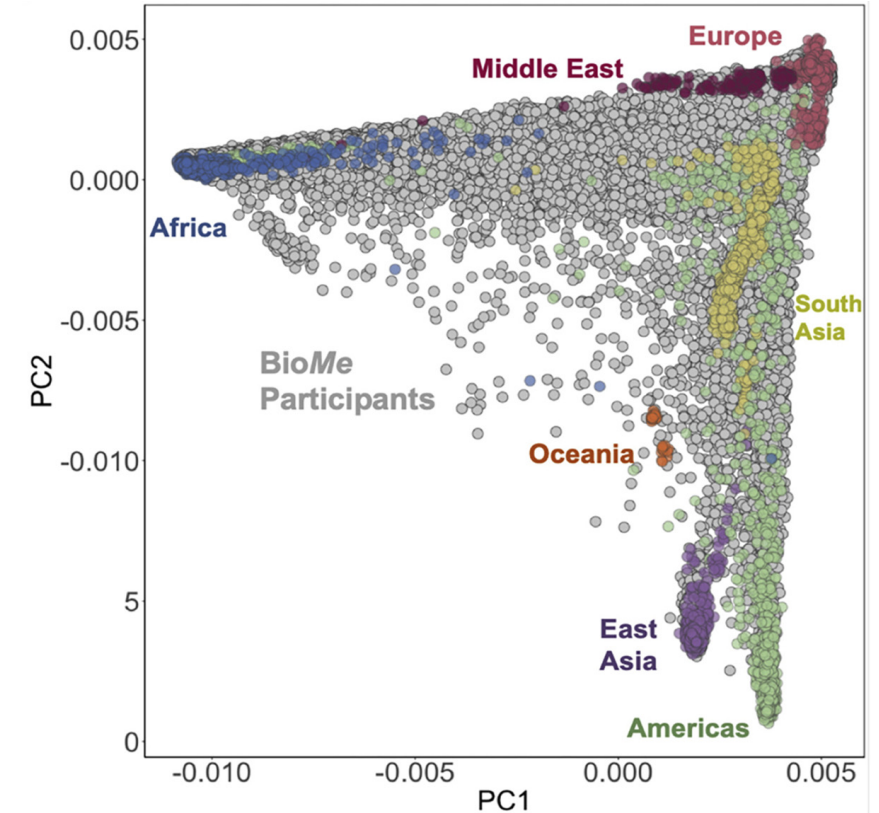
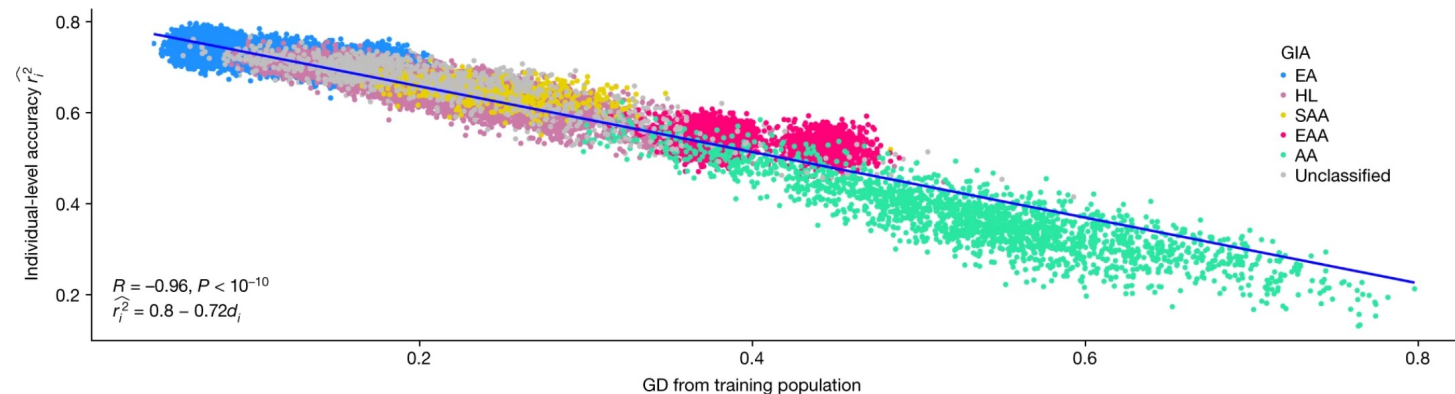
Genetic ancestry is continuous



Genetic ancestry is continuous

Impossible to define perfect boundaries

Modern approaches include calculating “genetic distance” from a training population



POP QUIZ

Which of the following do you believe to be true?

- A. Race is a proxy for genetic variation between populations.
- B. The term “Caucasian” is synonymous with European ancestry.
- C. Researchers should explain why they use population descriptors.
- D. The use of broad racial categories, e.g. “Asian,” is sufficient for human genetics research.
- E. None of the above are true.

POP QUIZ

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- C. Researchers should explain why they use population descriptors.
- ~~D. The use of broad racial categories, e.g. “Asian,” is sufficient for human genetics research.~~
- ~~E. None of the above are true.~~

How are population descriptors used in human genomics?

- Genetic ancestry
- Race
- Ethnicity
- Population group
- Genetic similarity

How are population descriptors used?

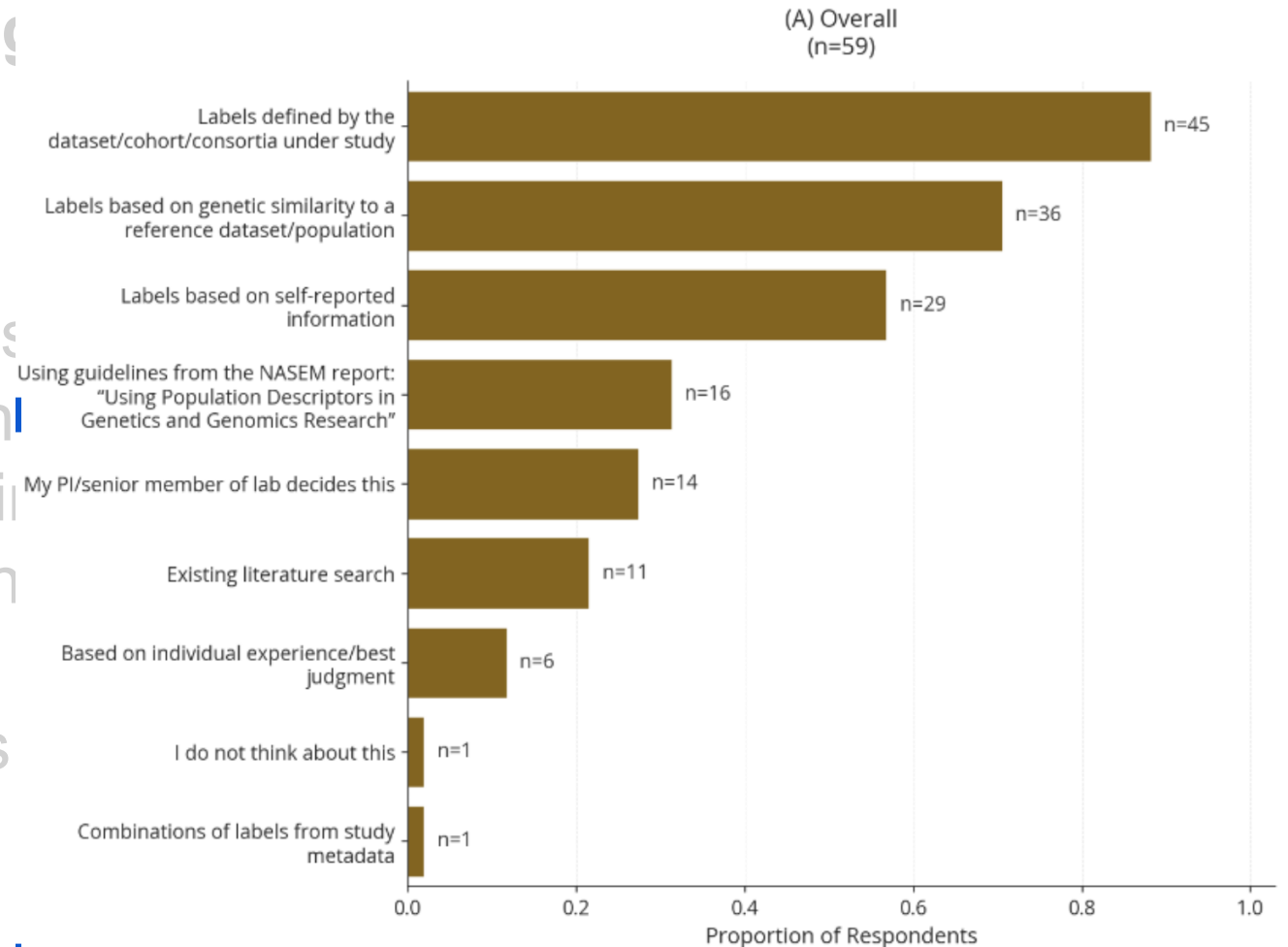
How do you currently assign population descriptors in your research?

- Existing literature search
- Labels defined by the dataset/cohort/consortium
- Labels based on participants' self-reported info
- Labels based on genetic similarity to a reference population
- Based on my own experience/judgment
- My PI decides this
- Using published guidelines
- I don't think about this.

How are population descriptors used?

How do you currently assign population descriptors in research?

- Existing literature search
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- Using published guidelines
- I don't think about this.



Existing and recommended practice

Commonly used in genetic research	Issue	NASEM-recommended
“African American” or “Black”	Conflates self-identified race and social identity with genetic ancestry	Keep race and ethnicity only as self-reported identity. Don’t use it to infer genetic ancestry
“African ancestry” or “European ancestry”	Implies discrete, bounded continental categories. But, genetic variation is continuous.	Genetic similarity to a specified reference panel (e.g. “1KG-AFR-like”)
“Caucasian”	Imprecise, outdated, and conflates geography and race	Avoid entirely. Use genetic similarity or specific geographic or ethnic self-identified labels
“Hispanic/Latino” (used to infer admixture)	Implies a discrete category but actually this label has little to do with genetics.	Self-identified ethnicity should be preserved separately. If inferring ancestry, use reference panel similarity.

Existing and recommended practice

What you should do depends on
your research question.

Self-identified labels are still scientifically useful
(e.g. for questions of health disparities and gene-environment interactions)!

Scenarios and recommendations

TABLE S-1 Recommended Approaches for the Use of Population Descriptors by Genomics Study Type

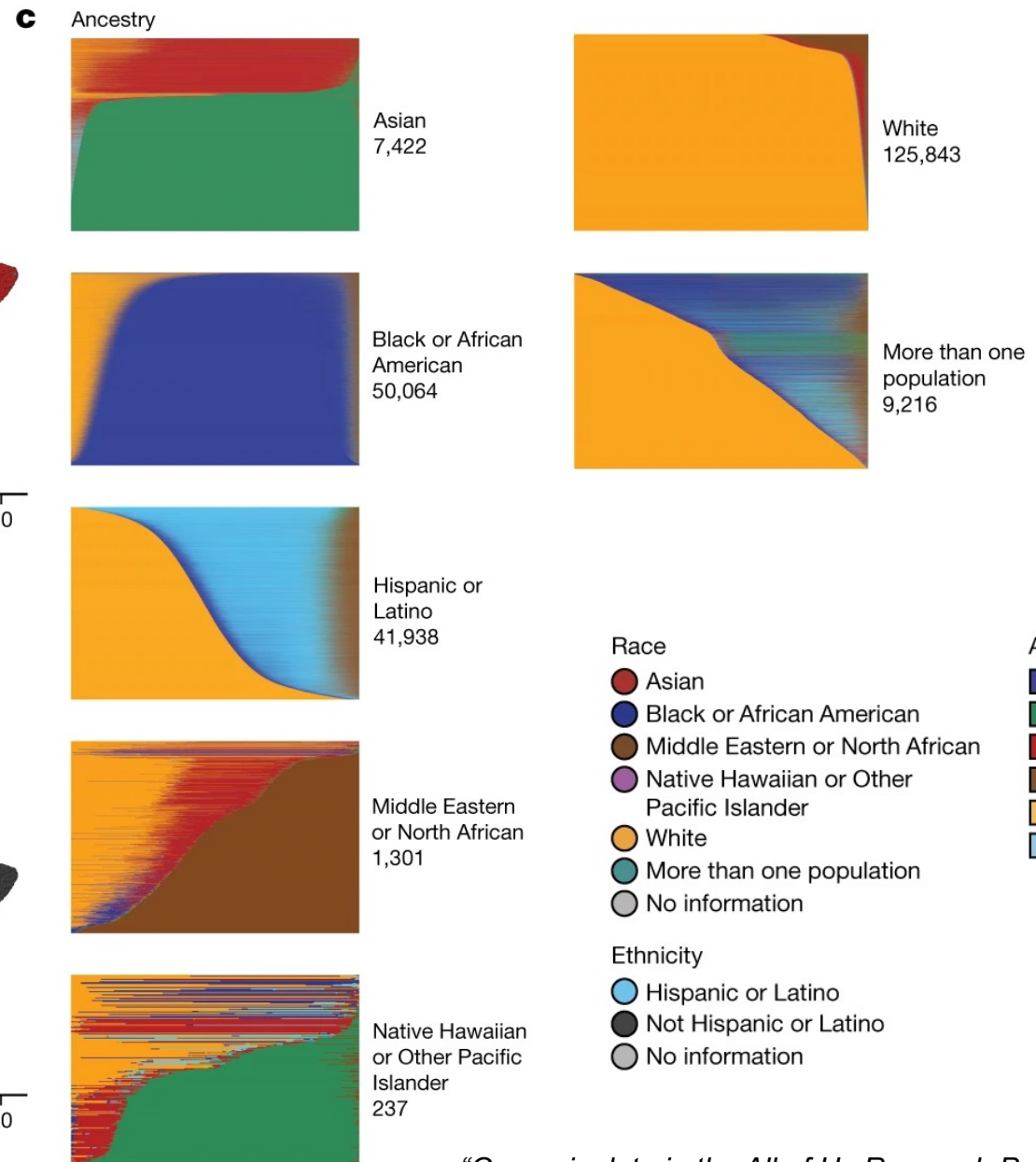
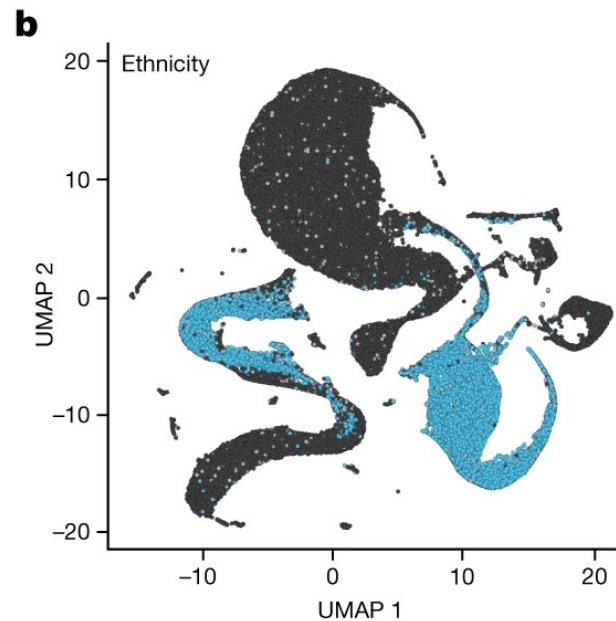
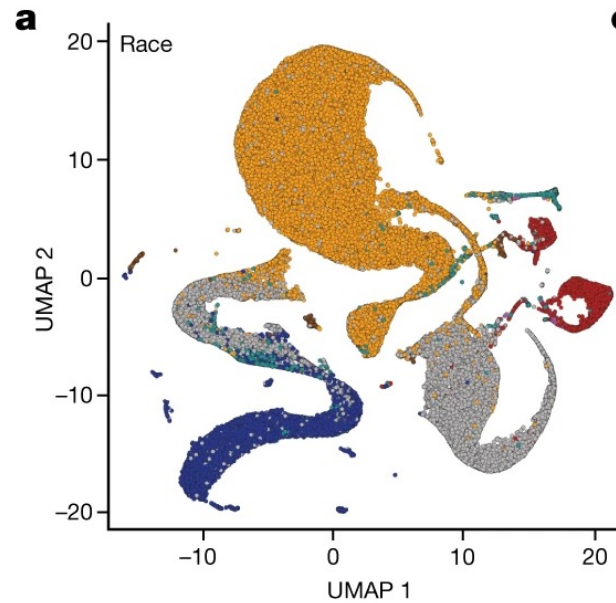
This table should be read and interpreted in conjunction with the report text. Consult the decision tree in Appendix D for more information and Chapter 5 text for best practices for each study type. See also the terminology box preceding the table and descriptions of each study type in Chapter 1 section “Classification of Genomics Study Types.” For any given study, the use of multiple descriptors may be preferable.						
LEGEND						
 Preferred population descriptor(s)				 Should not be used		
 In some cases; refer to Ch. 5 text and the decision tree in Appendix D				 Descriptors could be used if appropriate proxies for environmental, not genetic, effects		
GENOMICS STUDY TYPE	Race	Ethnicity/Indigeneity	Geography	Genetic Ancestry	Genetic Similarity	Notes
1: Gene Discovery – Mendelian Traits						Similarity suffices as a genetic measure; at fine-scale, other variables may be useful
2: Trait Prediction – Mendelian Traits						No population descriptors may be necessary for analysis
3: Gene Discovery – Complex Traits						Similarity suffices as a genetic measure
4: Trait Prediction – Complex Traits						Similarity suffices as a genetic measure
5: Cellular and Physiological Mechanisms						No population descriptors may be necessary for analysis
6: Health Disparities with Genomic Data						Not all health disparities studies rely on descent-associated population groupings, so none may be necessary for analysis
7: Human Evolutionary History						Reconstructing genetic ancestry may be of central interest

Scenarios and recommendations

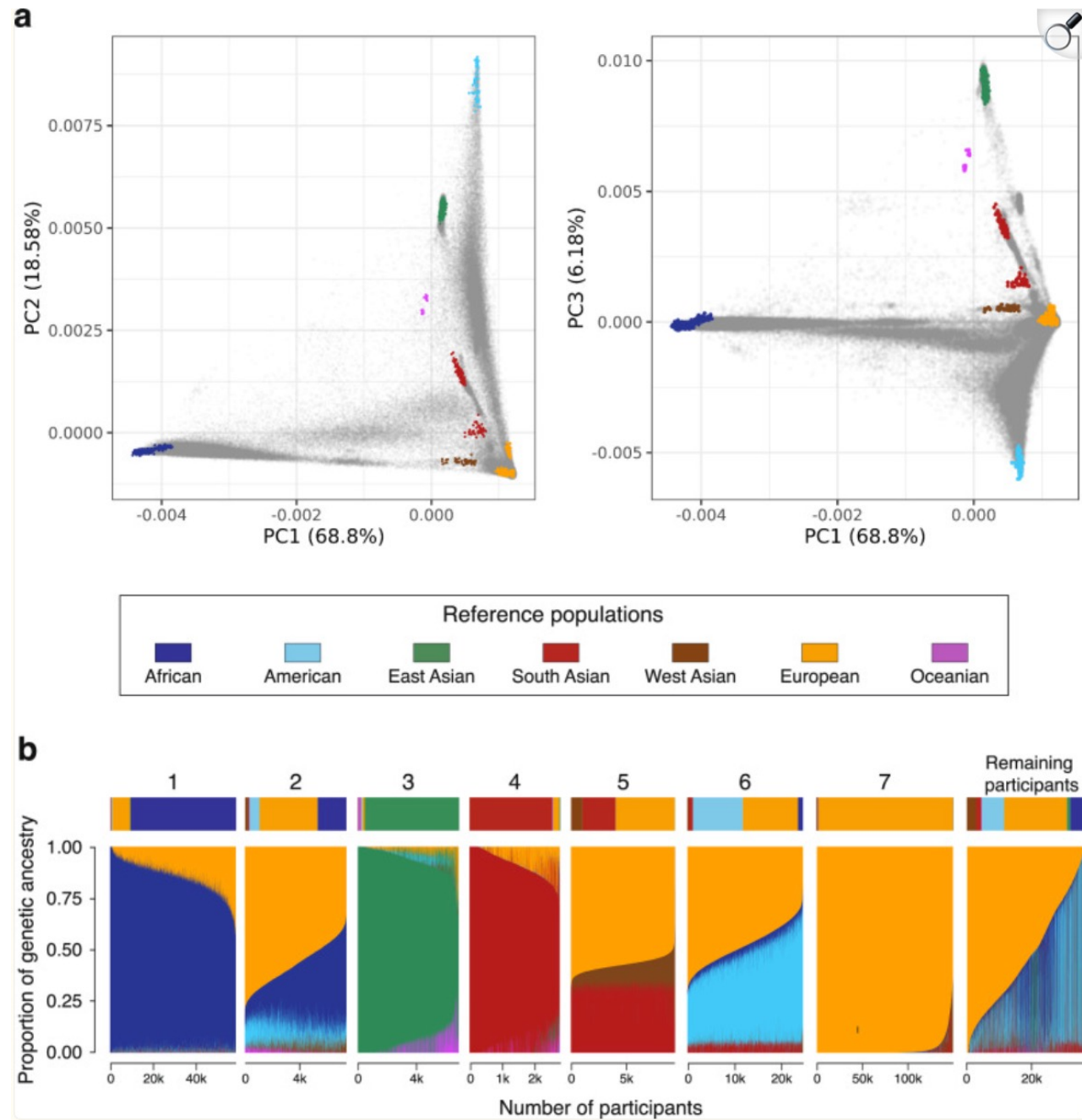
GENOMICS STUDY TYPE	Race	Ethnicity/Indigeneity	Geography	Genetic Ancestry	Genetic Similarity	Notes
1: Gene Discovery – Mendelian Traits	–	?	?	?	+	Similarity suffices as a genetic measure; at fine-scale, other variables may be useful
2: Trait Prediction – Mendelian Traits	–	E	E	?	+	No population descriptors may be necessary for analysis
4: Trait Prediction – Complex Traits	–	E	E	?	+	Similarity suffices as a genetic measure
6: Health Disparities with Genomic Data	E	E	E	?	+	Not all health disparities studies rely on descent-associated population groupings, so none may be necessary for analysis

Should not be used	–	+
Descriptors could be used if appropriate proxies for environmental, not genetic, effects	E	?
		In some cases; refer to Ch. 5 text and the decision tree in Appendix D

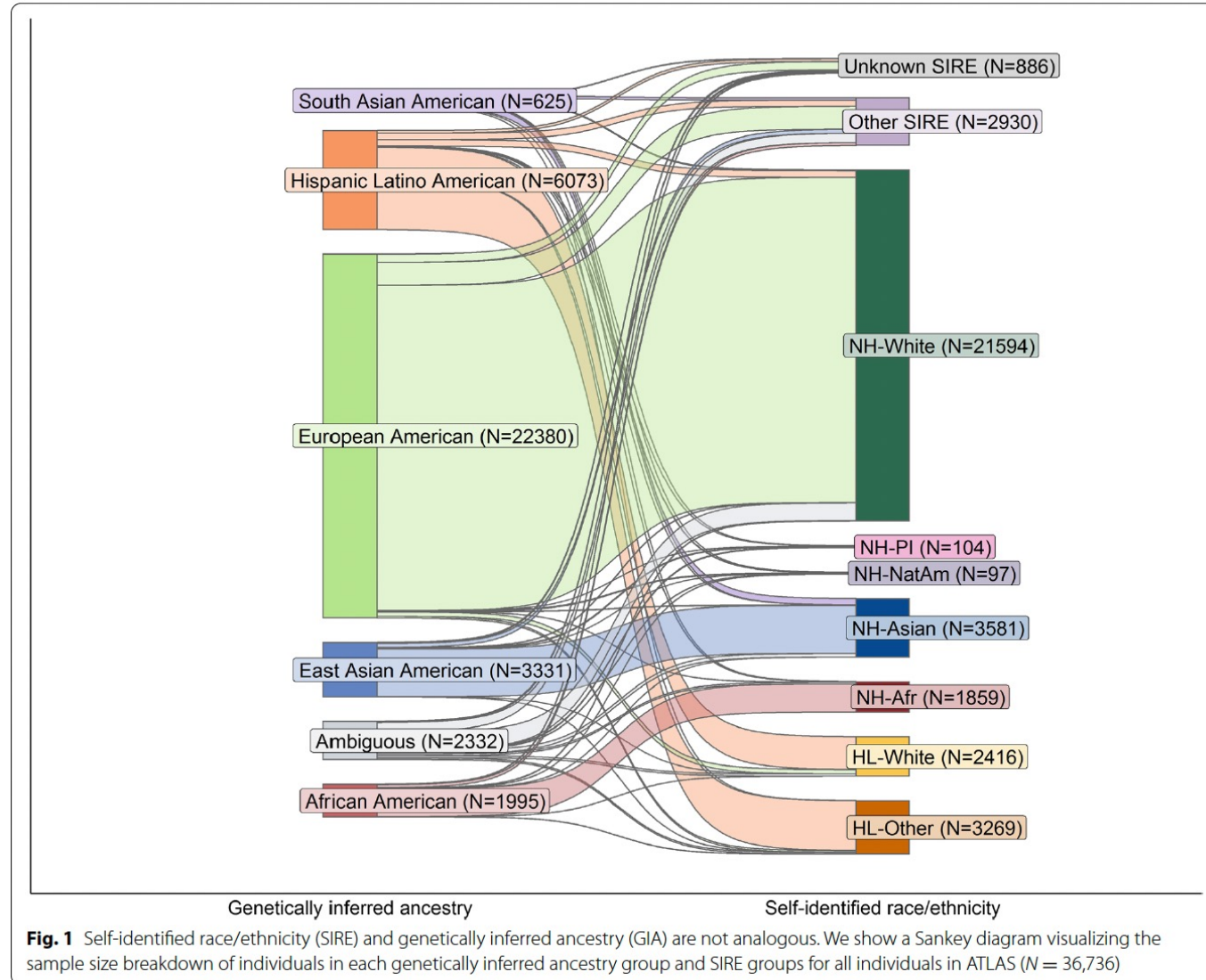
Not great



Much better!



Labels are messy!



Gene-environment interactions

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What is GxE?

Why study genes and environment together?



Leverage assumed effect
modifiers to increase power



Better characterize
biological mechanisms



Improve risk prediction and
prognostic models

Gene-environment interaction (GxE)

- “a *different effect* of an environmental exposure on disease risk in persons with different genotypes,”
or, equivalently,
- “a *different effect* of a genotype on disease risk in persons with different environmental exposures.”

Three kinds of interaction

Biologic
interactions

Public health
interactions

Statistical
interactions

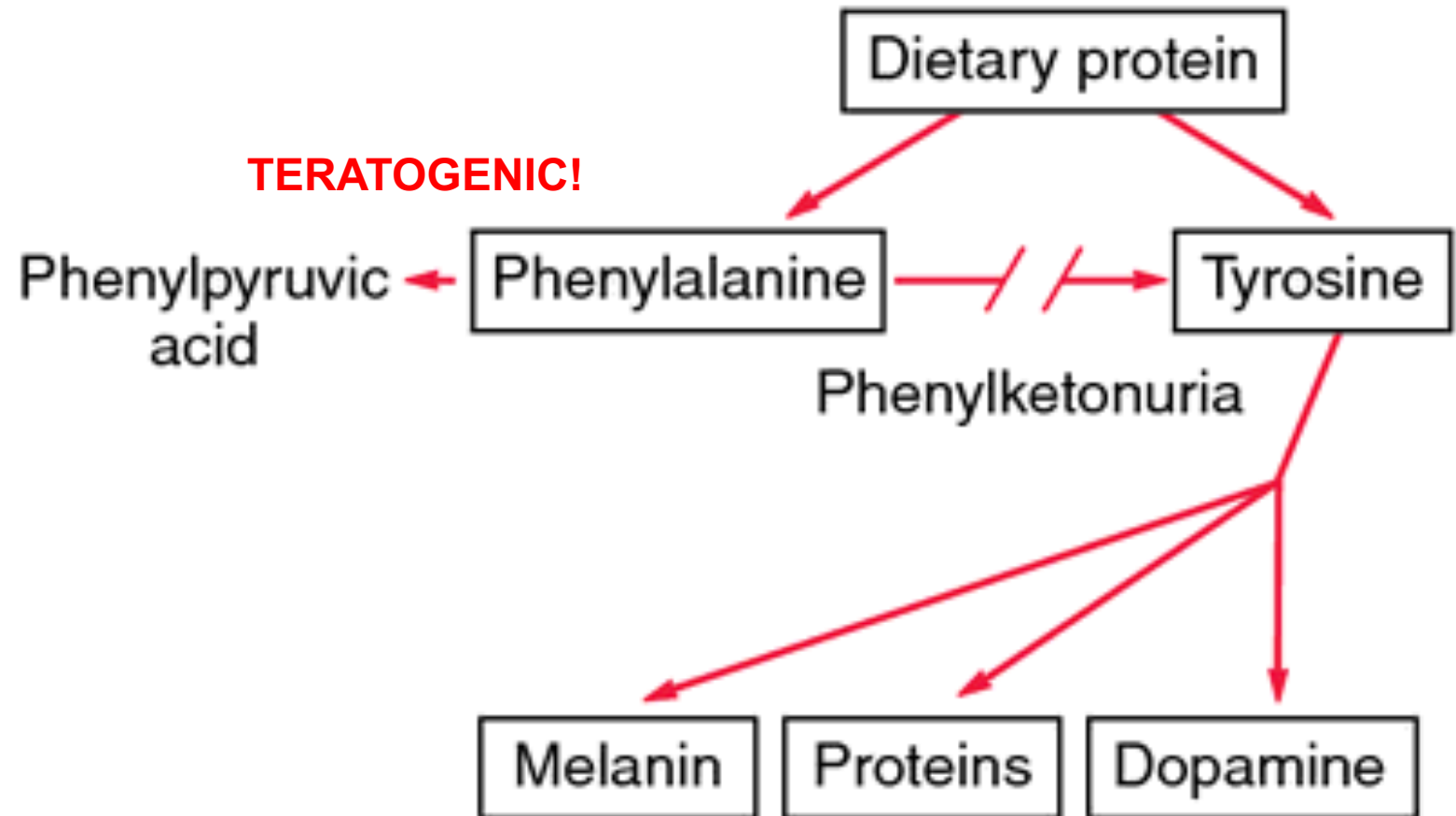
Biologic interactions

- “necessity of a causal partner for disease occurrence”
- “occurs when one risk factor allows the other to be expressed in a disease outcome.”
- hard to define...

The GxE Poster Child: PKU

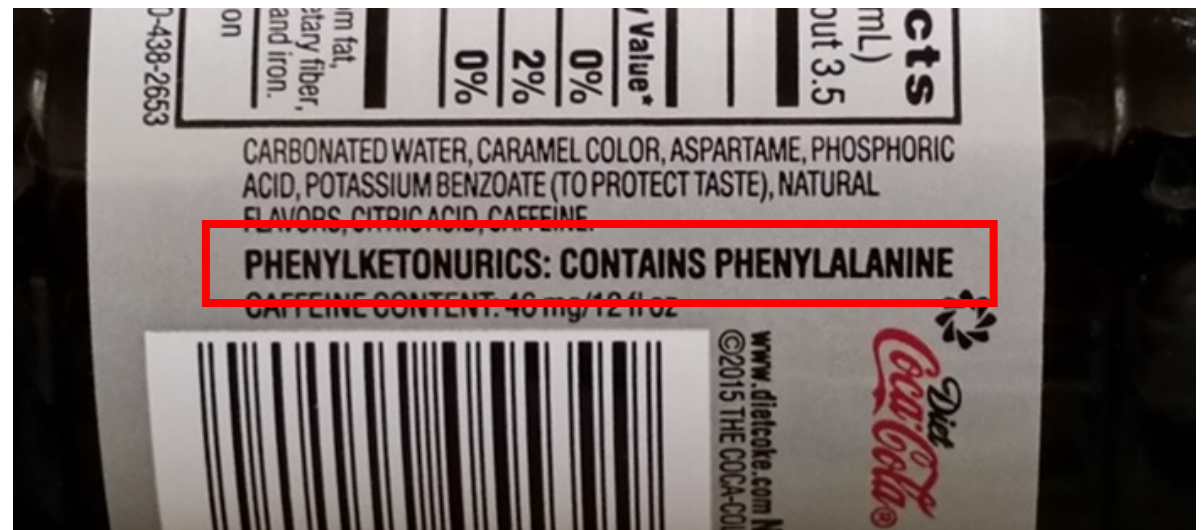
PKU occurs due to defects in the PAH gene that creates an enzyme to break down phenylalanine, which leads to buildup of toxic phenylalanine.

Autosomal recessive mode of inheritance



The GxE Poster Child: PKU

- Neither genotype nor exposure cause disease *alone*.
- Only variant genotype and normal dietary phenylalanine exposure together produce harmful outcomes.
- Penetrance is biologically/mechanistically consistent.
- Preventive intervention exists and is relevant specifically to carriers!



Sufficient Component Cause/ Counterfactual Interaction

“[W]e may be interested whether, for some individuals, an outcome occurs if both of two exposures are present but not if only one or the other is present.”

Under strong assumptions, a specific form of statistical interaction—departure from additivity on the absolute risk scale—implies interaction in this sense.

Three kinds of interaction

Biologic
interactions

Public health
interactions

Statistical
interactions

Public health interactions

- “situation in which the public health costs or benefits from altering one factor must take into account the prevalence of other factors”
- e.g. carriers of some allele A may benefit disproportionately from some intervention
- Assuming the public health benefit has to do with reducing disease incidence, this corresponds to departures from additivity on the absolute risk scale.

Association of Aspirin and NSAID Use With Risk of Colorectal Cancer According to Genetic Variants

- Motivation: Aspirin/NSAID use is associated with lower risk of colorectal cancer
- Methods: Case-control study with (1) logistic regression including interaction and (2) case-only interaction analysis
- Aspirin/NSAID use was protective for individuals with TT genotype at rs2965667, but led to higher risk among those with rarer TA or AA genotypes.
- Public health benefit is applicable in specific subsets of the population.

Three kinds of interaction

Biologic
interactions

Public health
interactions

Statistical
interactions

Statistical interactions

- “variation in some measure of the effect of an exposure on disease risks across the levels of ... a modifier. ... The definition of interaction depends on the measure of association used.
- Departure from an additive effects model on a particular scale

Example of statistical interaction

$$G \begin{cases} 1 \text{ if carrier} \\ 0 \text{ if non-carrier} \end{cases} \quad E \begin{cases} 1 \text{ if exposed} \\ 0 \text{ if unexposed} \end{cases}$$

Risk of disease

$$p_{GE} = b_0 + b_g G + b_e E + b_{ge} GE$$

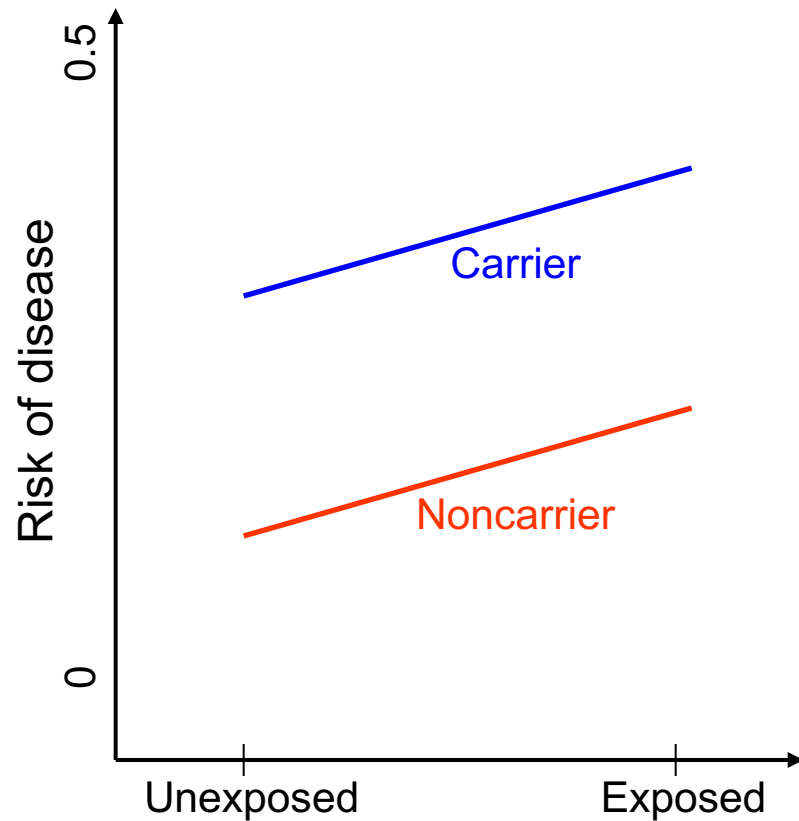
Log odds of disease

$$\log \frac{p_{GE}}{1-p_{GE}} = \beta_0 + \beta_g G + \beta_e E + \beta_{ge} GE$$

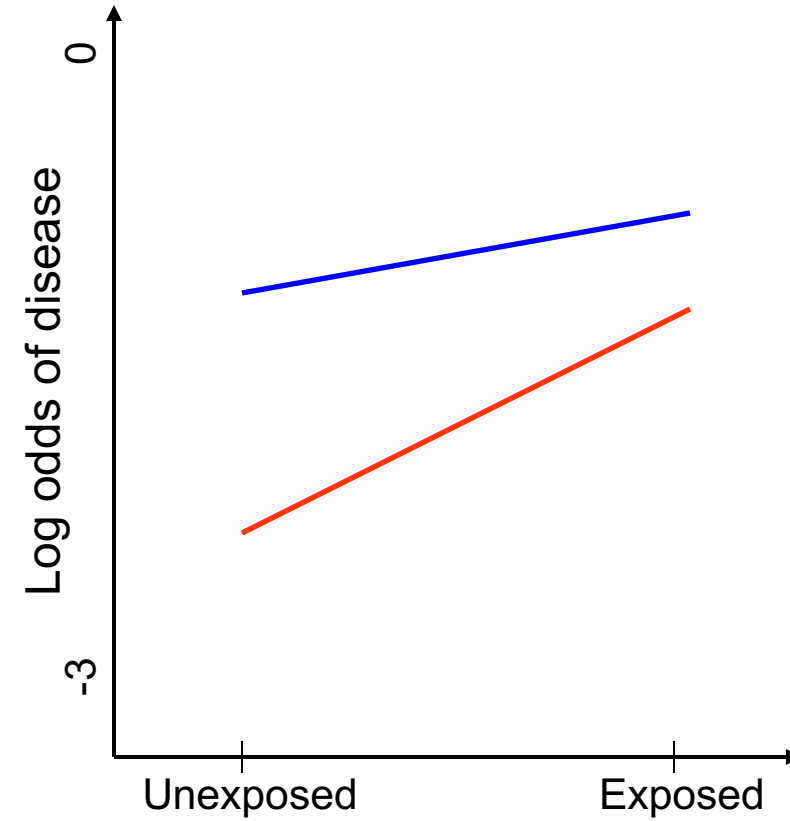
Test for “additive interaction:” H_0 is $b_{ge}=0$

Test for “(multiplicative) interaction:” H_0 is $\beta_{ge}=0$

Interaction scale

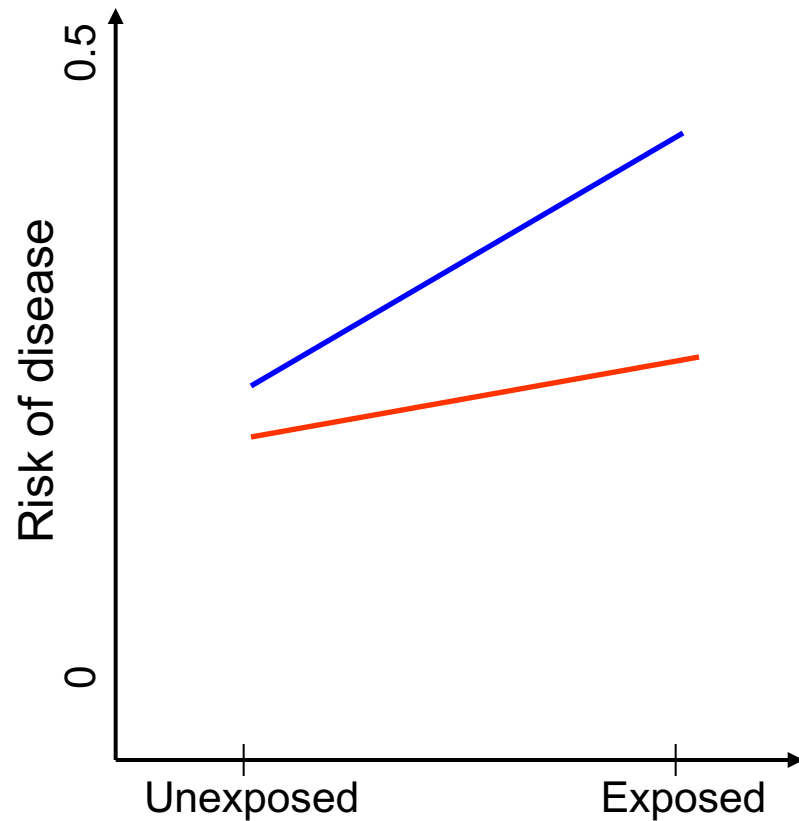


$$b_{ge}=0$$

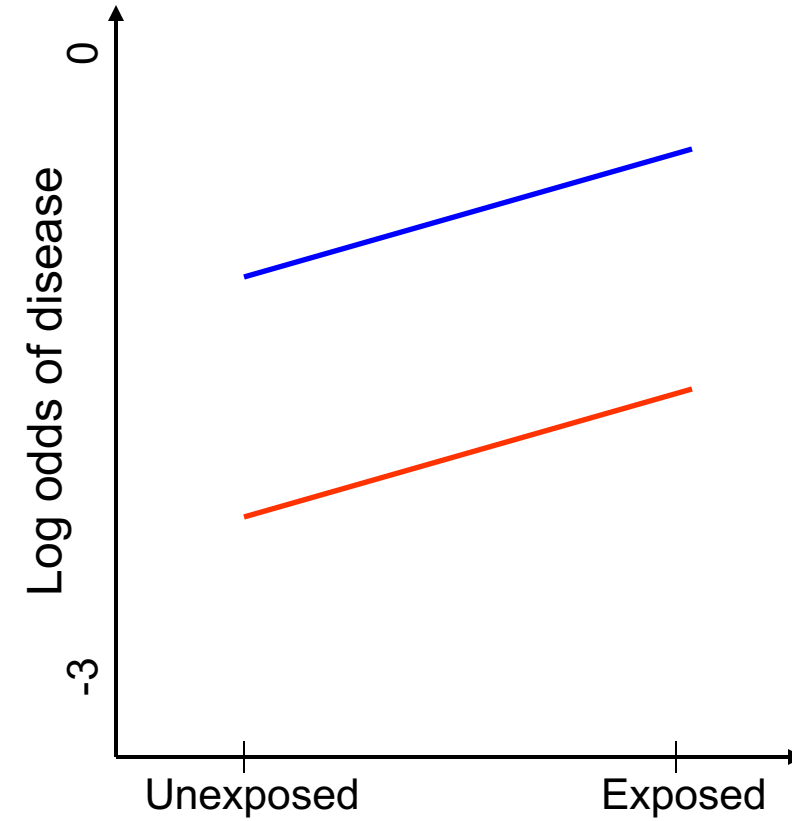


$$\beta_{ge} \neq 0$$

Interaction scale matters



$$b_{ge} \neq 0$$



$$\beta_{ge} = 0$$

“Types” of GxE

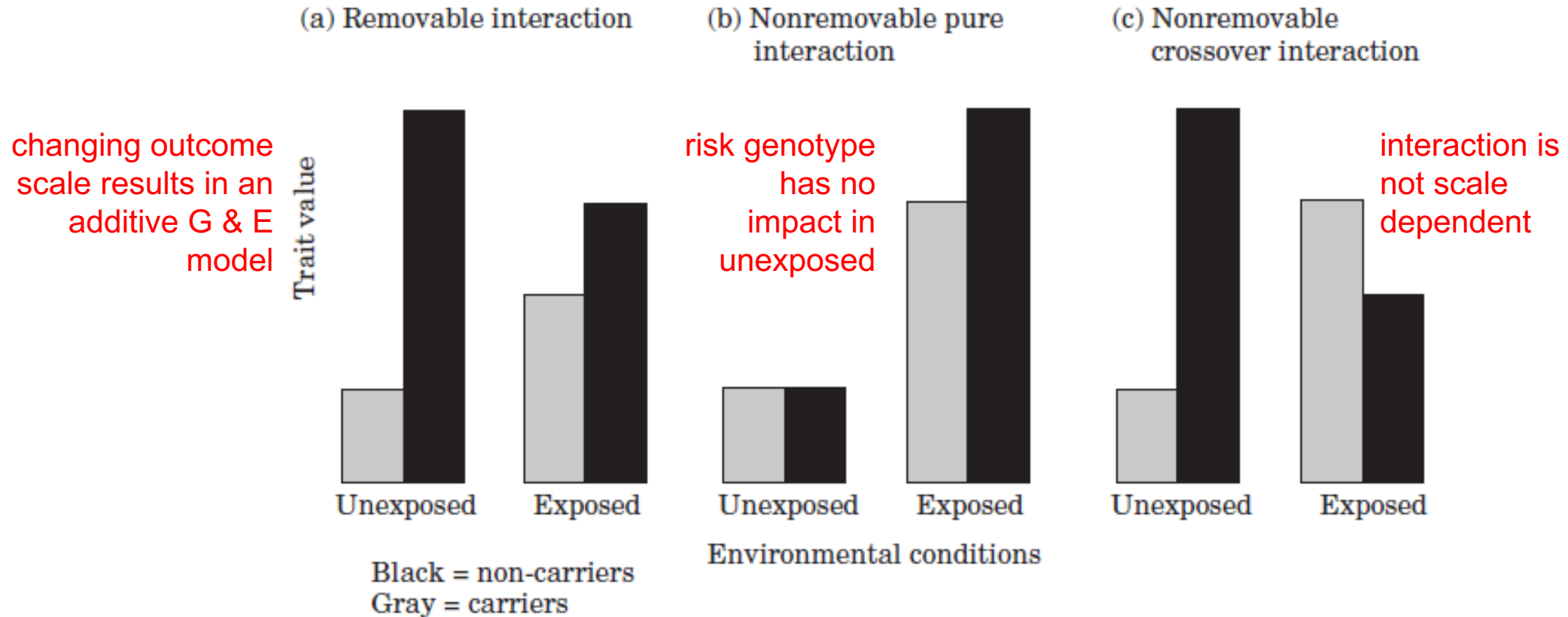
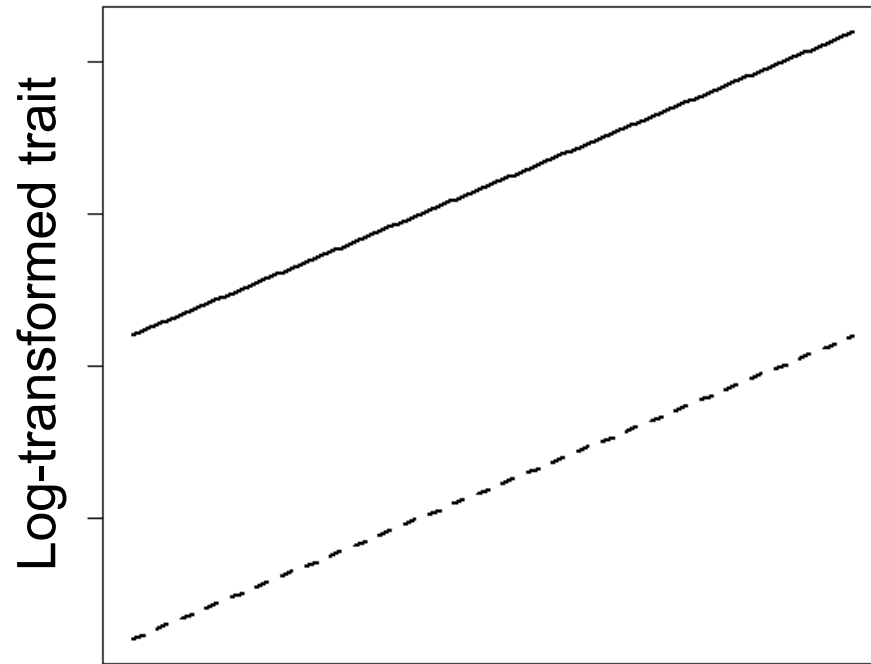
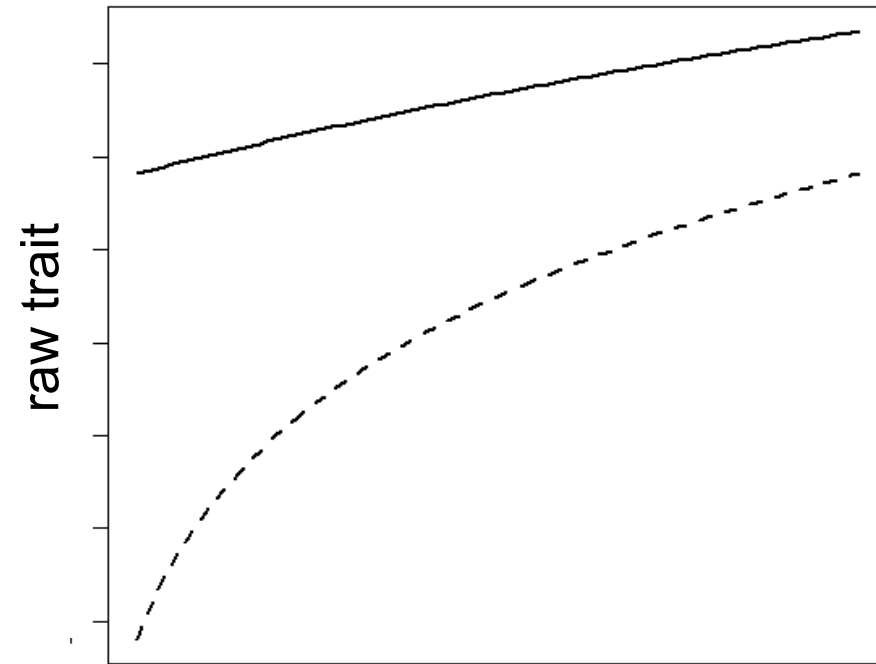


Figure 9.2 Three qualitative patterns of gene–environment interaction. The y-axis represents a trait value (e.g., mean height, disease prevalence, expected survival); the x-axis represents two environmental conditions; the black bars denote noncarriers and the gray bars represent carriers. (a) gives an example of a removable interaction. (b) and (c) are nonremovable interactions (“pure” and “crossover” interactions, respectively).

Outcome scale



E
 $Y = \beta_G G + \beta_E E + \text{error}$
fits well

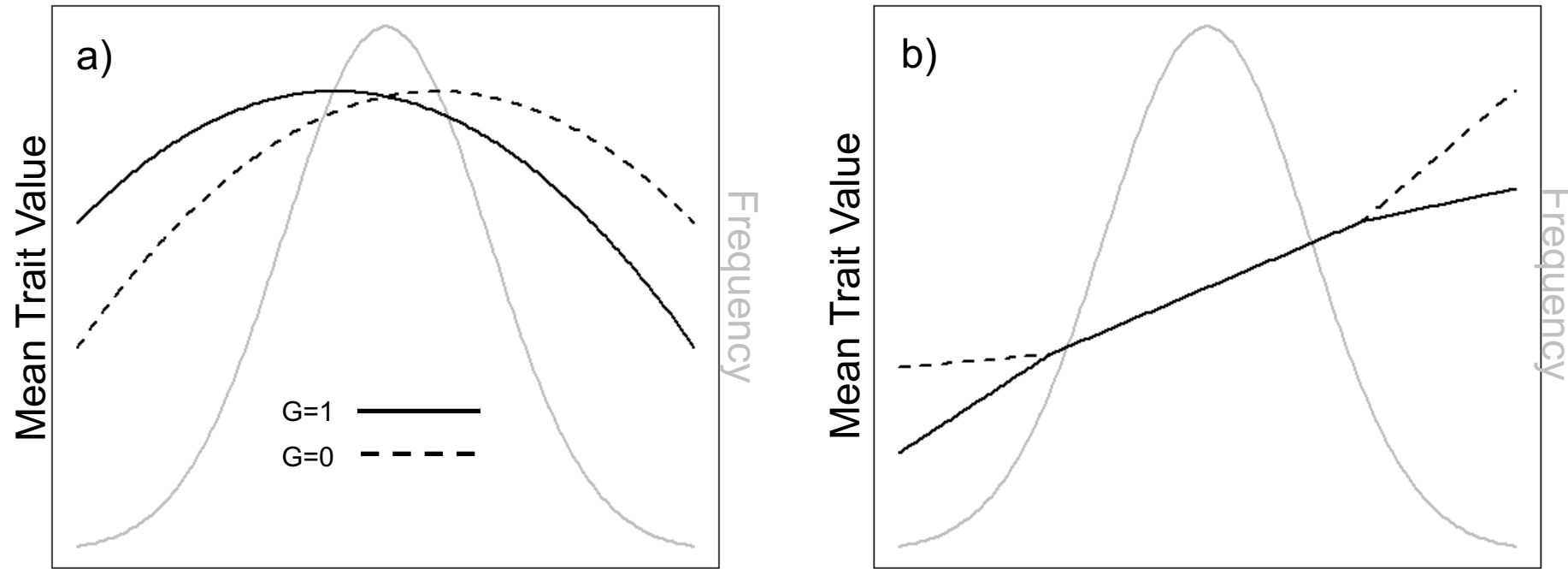


E
 $Y = \beta_G G + \beta_E E + \text{error}$
doesn't fit well

Gene-environment interactions can be easily created or eliminated by changing the scale of Y. There is no universally appropriate scale.

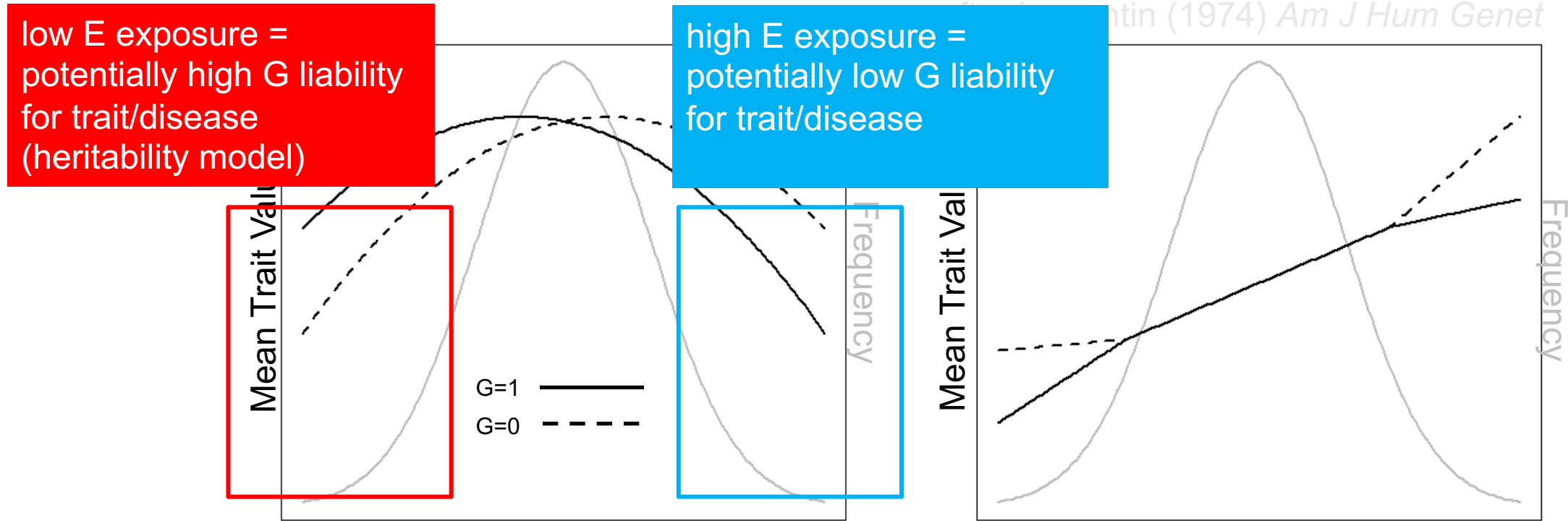
Range of E exposures

Two “reaction norms” (i.e. gene-environment interaction patterns)
after Lewontin (1974) *Am J Hum Genet*



Genetic effect, environmental effect, and gene-environment interaction all depend on **what part of the E distribution you've sampled**: this has implications for discovery and replication

Range of E exposures



Genetic effect, environmental effect, and gene-environment interaction all depend on **what part of the E distribution you've sampled**: this has implications for discovery and replication

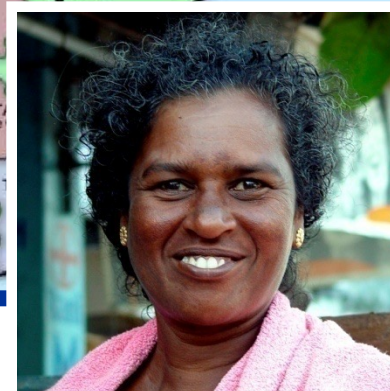
India health study



New Delhi



Trivandrum



Participant characteristics by region

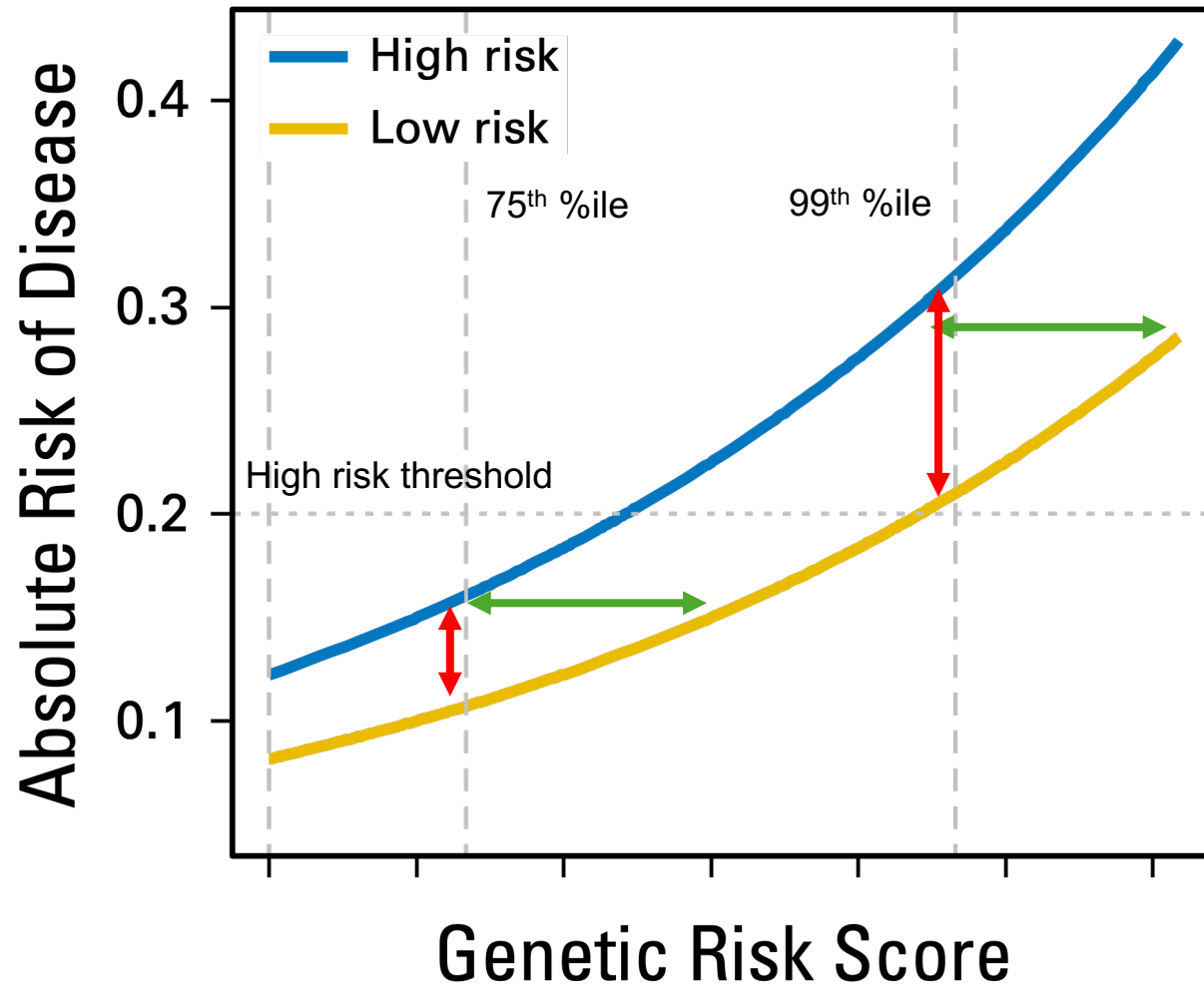
Characteristic	New Delhi	Trivandrum
Total (n=1,313)	n=619	n=694
Age, years (mean, SD)	47.4 ± 10.0	48.8 ± 9.2
Household monthly income, %		
<5,000 rupees	7.1	71.9
>10,000 rupees	76.7	3.1
Household items, %		
Car	25	7
Refrigerator	87	58
Washing machine	79	14
Total physical activity, MET-hr/wk	42.5 ± 43.8	147.3 ± 85.2
Vigorous physical activity, MET-hr/wk	0.6 ± 6.8	26.2 ± 51.4
Sitting, hr/day	10.4 ± 2.0	5.0 ± 2.3
Centrally obese, %	82.1	60.2

Association of *FTO* rs3751812 with waist circumference

Characteristic	N	Effect size per T allele (95% CI)	P _{trend}	
Overall	1,209	+1.61 cm (0.67, 2.55)	0.0008	0.009
New Delhi				
Overall	578	+2.53 cm (1.08, 3.97)	0.0006	0.59
By PA				
≤ 91 MET-hrs/wk	517	+2.36 cm (0.82, 3.89)	0.003	
92-151 MET-hrs/wk	32	+6.39 cm (1.94, 10.85)	0.005	
152-217 MET-hrs/wk	24	-0.95 cm (-7.33, 5.42)	0.77	
218+ MET-hrs/wk	5	N/A	N/A	
Trivandrum				
Overall	574	+0.87 cm (-0.35, 2.08)	0.16	0.004
By PA				
≤ 91 MET-hrs/wk	170	+3.50 cm (0.90, 6.10)	0.008	
92-151 MET-hrs/wk	132	+1.13 cm (-1.08, 3.33)	0.32	
152-217 MET-hrs/wk	141	+1.04 cm (-1.63, 3.70)	0.45	
218+ MET-hrs/wk	131	-2.32 cm (-4.82, 0.18)	0.07	



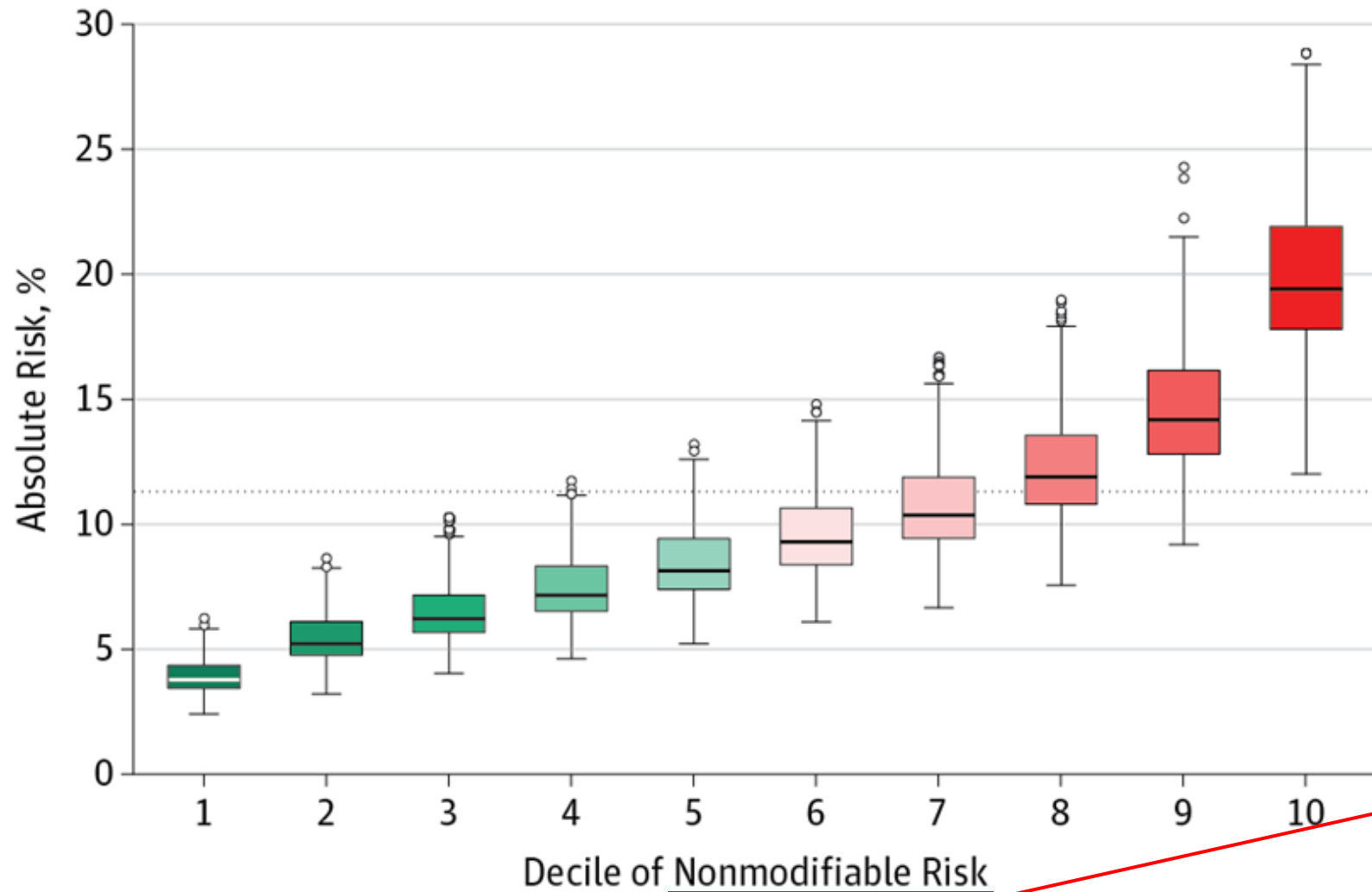
Statistical interaction $\neq$ Public health interaction



Even though there's **technically no multiplicative interaction here**, the **absolute** gap between the high and low risk curves is much wider at higher levels of genetic risk.

1. Individuals at the upper tail of genetic risk will disproportionately benefit from some intervention.
2. Those at the tail of unexposed individuals can avoid any risks from some intervention.

Absolute risk improvements greatest in highest risk women with breast cancer



Women in the highest decile of non-modifiable risk (mostly due to genetics) can reduce their absolute risk for breast cancer through healthy lifestyle choices much more than those already at low genetic risk.

SNPs, family history, reproductive history, height

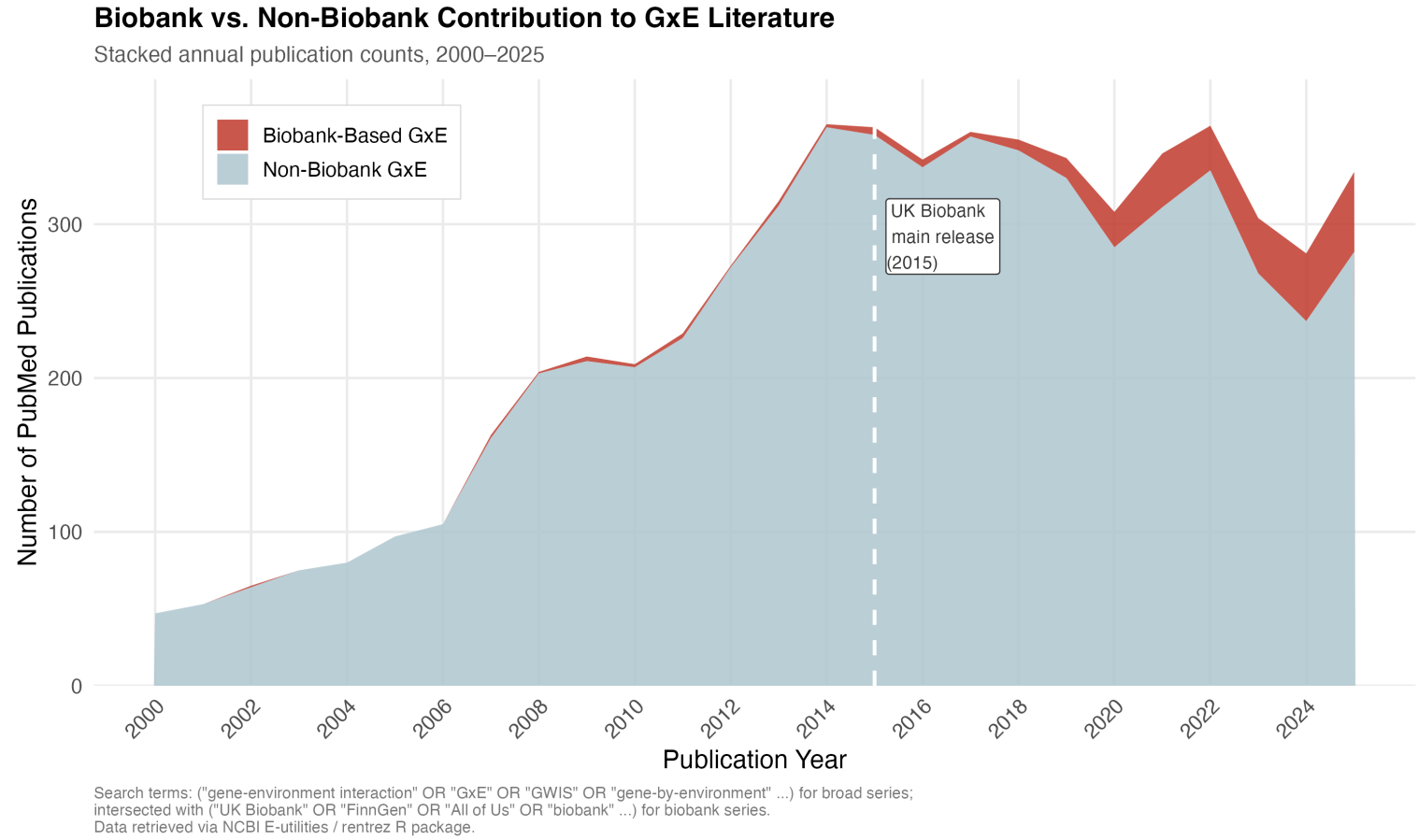
So, what do we mean when we say GxE?

- Statistical or public health interactions
- Most often, **statistical interaction**

State of the science

GxE literature has grown over time

- Originally, interactions of singular exposures with candidate genes
- Now, large-scale population GxE is more common



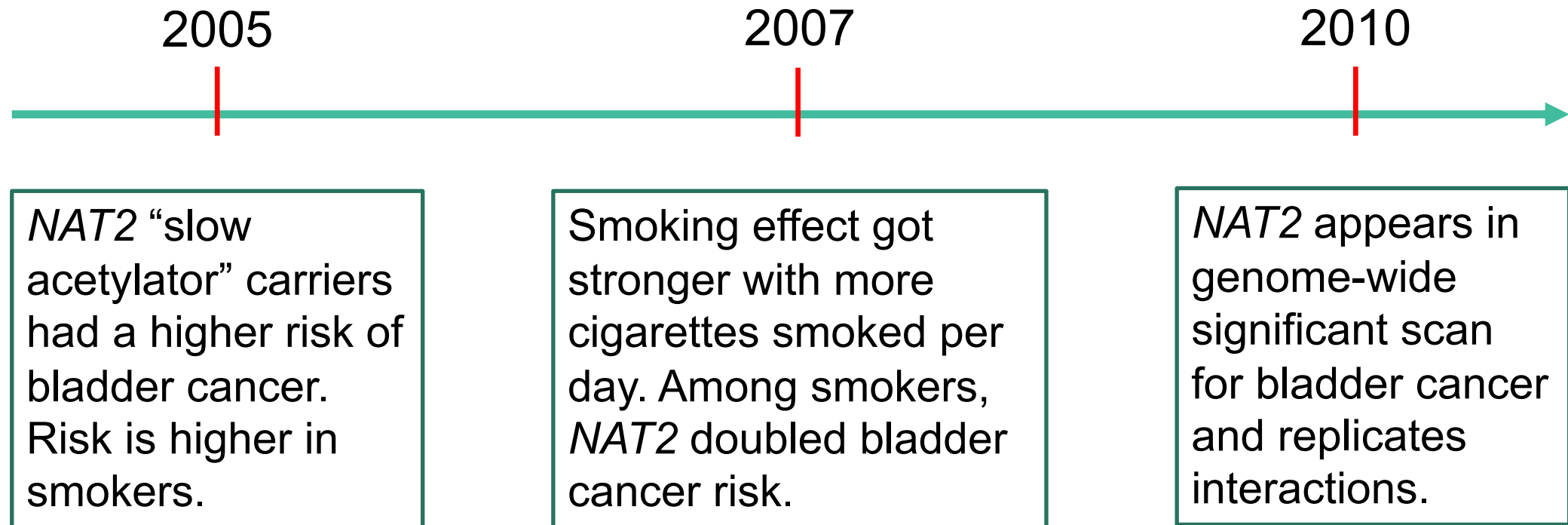
A majority of these studies report a significant gene-environment interaction.

Are these all credible?

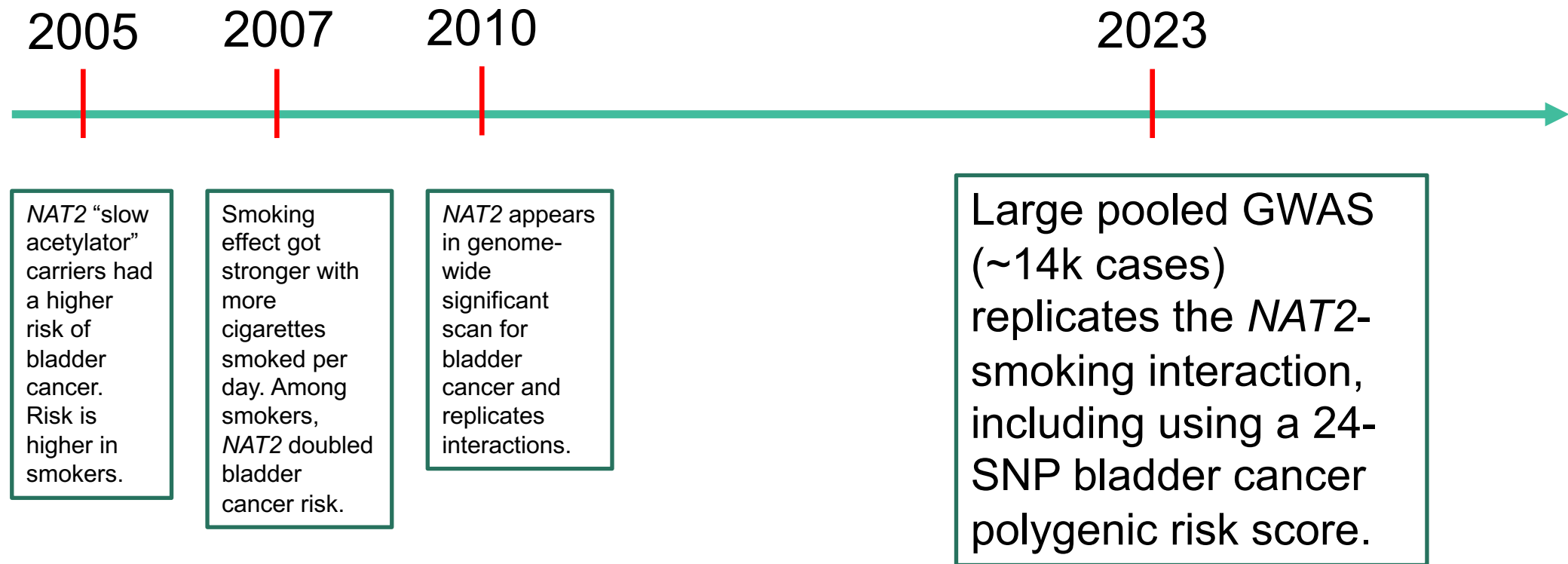
Perhaps not.

1. Small sample sizes.
2. Limited correction for multiple testing/low priors.
3. Limited replication.

NAT2, smoking, and bladder cancer



NAT2, smoking, and bladder cancer



ALDH2, alcohol, and esophageal cancer

2001

Inactive *ALDH2* variant carriers clear acetaldehyde much slower. Identifies higher multiplicative risk of esophageal cancer (EC) for drinking.

2010

Carrying risk alleles at *ALDH2* and *ADH1B* compounds EC risk further. Also identifies strong interaction with smoking behavior.

2011

ALDH2 variants appear in genome-wide significant scan for EC and replicates interactions with drinking & smoking.

Is a genetic association
actually a GxE interaction?

CHRNA3/4/5, smoking, and lung cancer

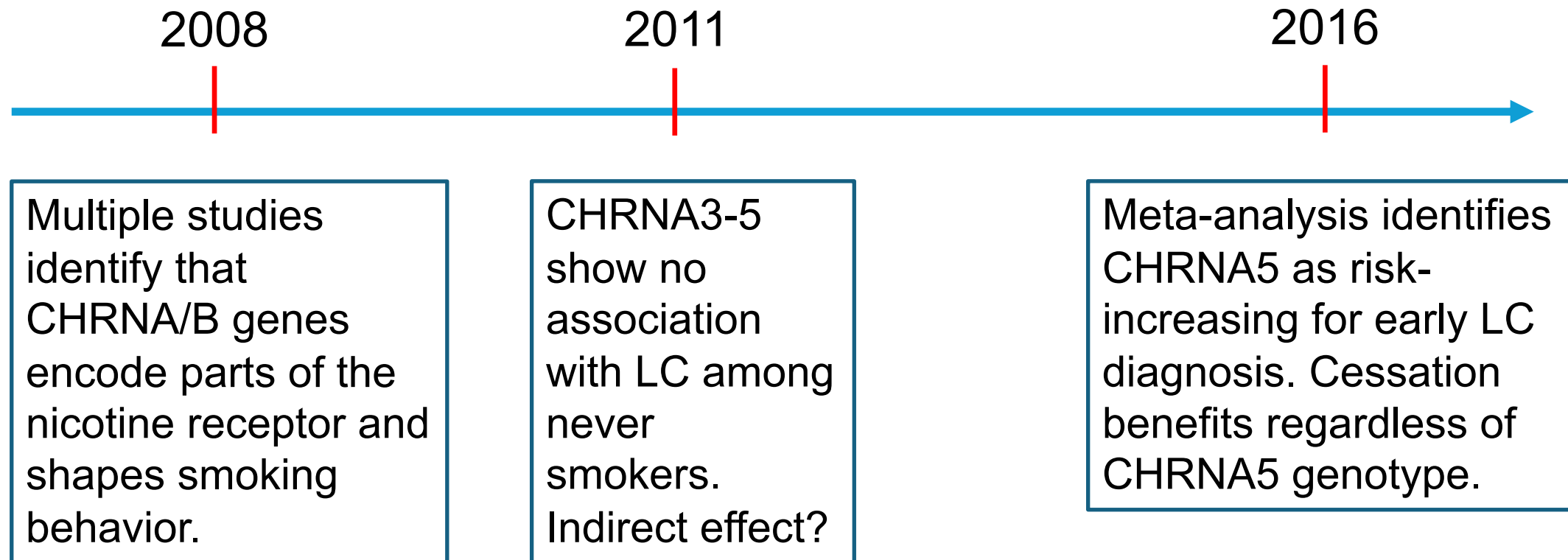


Table 1. Major findings on gene–environment interactions associated with human cancers.

Cancers Site/Types of Study	Genes (SNPs, Genotype, RS-Number, Chromosome, Allele) of Interest	Environmental Exposures	Key Findings on Gene–Environment Interactions Based on OR, RR, and/or HR
Lung cancer (LC) case-control studies [53]	miR-26a-rs7372209 (TT, CT, AA)	Regular exposure to cooking oil fume	TT and CT genotype carriers of miR-26a-1 rs7372209 and miR-16-1 rs1022960 who had been exposed to cooking oil fumes had a higher risk of developing LC compared to those who were not exposed to cooking oil fumes (OR = 1.743, 95% CI = 1.038–2.753, $p = 0.036$) and (OR = 2.326, 95% CI = 1.409–3.843, $p = 0.001$), respectively.
	miR-605 rs2043556		
	miR-16-1 rs1022960		
Lung cancer (LC) hospital-based case-control studies [54]	rs2608053 (AA, GG, AA/GG)	Heavy cigarette smoking	AG or GG genotype of rs151927 smoking were 7.18 times at risk of LC compared to those with AA genotype with no smoking exposure (OR = 7.108, 95% CI = 2.36–21.37). AG/GG genotypes of rs2608053 and rs13254990 polymorphism were more likely to develop lung cancer compared to non-smokers.
	rs1561927 (AA, GG, AA/GG)		
	rs13254990 (AA, GG, AA/GG)		
Colon cancer case-only and case-control studies from GWAS [55]	rs10849432 at 12p13.31	Alcohol consumption, lack of physical activity, use of aspirin, high intake of red and processed meat	SNP rs4444235 at 14q22.2 and regular exercise and the SNP rs2423279 at 20p12.3 and regular aspirin use may increase the risk of colorectal cancer initiation.
	rs11196172 at 10q25.2		
	rs4444235 at 14q22.2		
	rs2423279 at 20p12.3		
	rs2241714 at 19q13.2		
	rs1957636 at 14q22.3		

Cancers Site/Types of Study	Genes (SNPs, Genotype, RS-Number, Chromosome, Allele) of Interest	Environmental Exposures	Key Findings on Gene–Environment Interactions Based on OR, RR, and/or HR
Bladder cancer case-control studies [56]	rs1695 AA; GSTP1 AG Ile105Va GG	Environmental exposure to arsenic	Genotype–arsenic interactions in the high exposure group were linked to increased risk of bladder cancer initiation.
Breast (BC) case-control studies [57]	rs4808801 located on chromosome 19 rs11242675 located on chromosome 6 rs941764 located in chromosome 14 rs12422552 located in chromosome 12 rs16857609 located in chromosome 2	Age at first birth and menarche, number of pregnancies, obesity, smoking, and heavy alcohol drinking	Strong association was found between BC risk, rs4808801, and four and more full-term pregnancies (OR = 5.08, 95% CI = 0.77–0.93, $p = 2.0 \times 10^{-4}$). Interactions were also found between smoking, rs 11242675, and BC risk (OR = 1.13, 95% CI = 1.06–1.21, $p = 3.4 \times 10^{-4}$), and also between alcohol consumption, BC risk, and rs941764.
Prostate cancer case-only [58]	CC, TT, CT of JAZF1 genotype	Environmental exposure to lead and cadmium, and heavy cigarette smoking	Strong G × E interactions with carrier of CC genotype, exposure to Pb in houses built before 1960, and PC risk compared to CT or TT genotype (OR = 1.81, 95% CI = 1.04–3.16; $p = 0.036$).

OR: odds ratio; RR: relative risk; HR: hazard risk; CI: confidence intervals; MHT: menopausal hormone therapy; BMI: body mass index; GWAS: genome-wide association studies.

Polygenic score-environment interactions

Genome-wide composite measures can potentially capture more variance than single genes (in some cases).

Using a PGS instead of a candidate gene may present new findings/public health potential

- e.g. interactions between a 77-SNP breast cancer PGS and various environmental factors (BMI, alcohol use, etc.)



sociological science

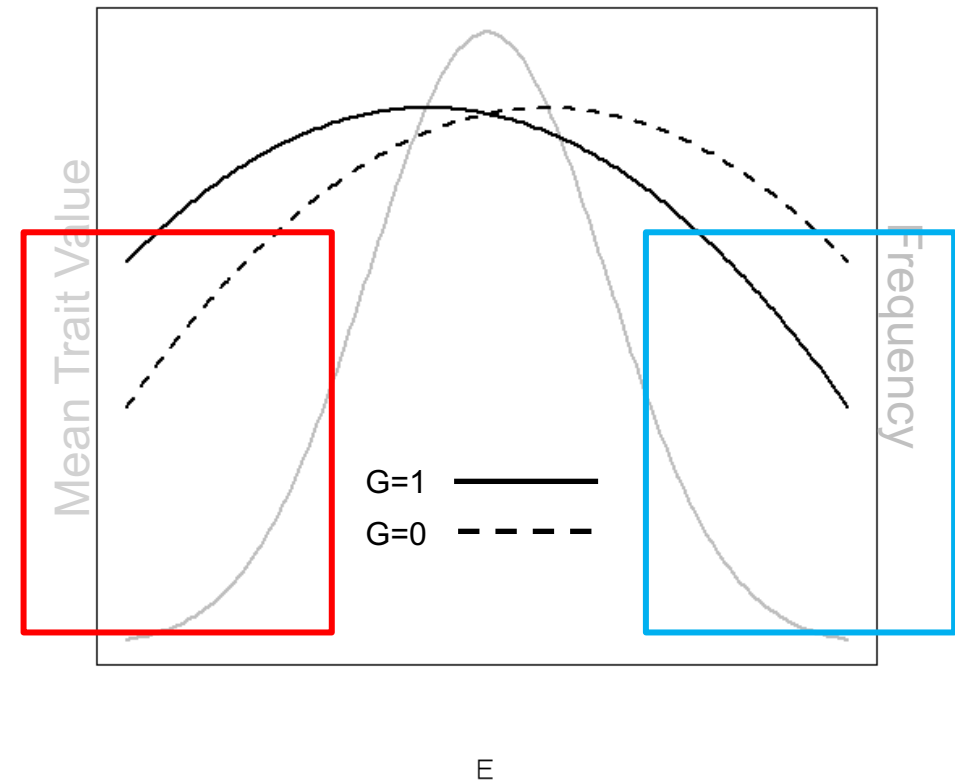
Interactions between Polygenic Scores and
Environments: Methodological and
Conceptual Challenges

Benjamin W. Domingue,^a Sam Trejo,^b Emma Armstrong-Carter,^a Elliot M.
Tucker-Drob^c

Challenges of GxE studies

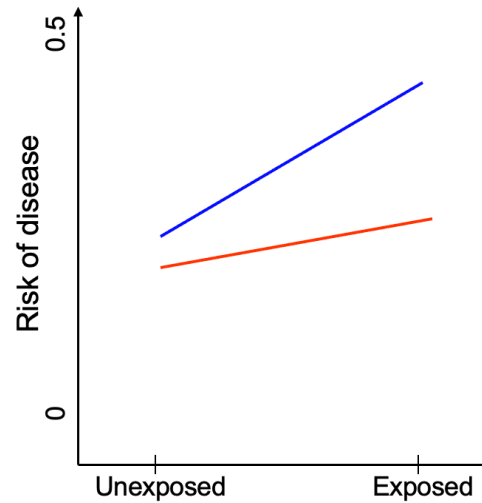
Common epi concerns

- Sampling biases
- Statistical and methodological artifacts
- Environmental measurements

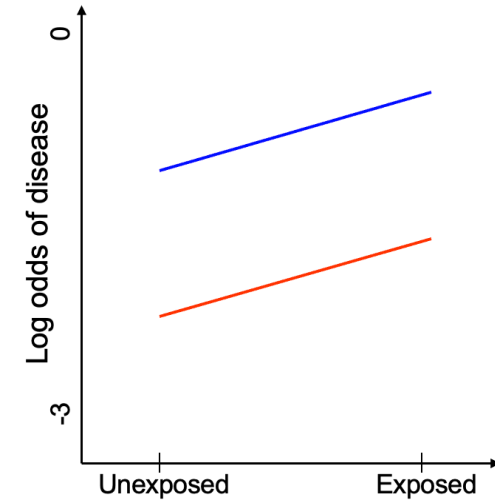


Common epi concerns

- Sampling biases
- Statistical and methodological artifacts
- Environmental measurements



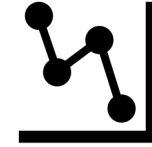
$b_{ge} \neq 0$



$\beta_{ge} = 0$

Common epi concerns

- Sampling biases
- Statistical and methodological artifacts
- Environmental measurements



“Environmental” measurements

Limited exposure range

- possible for some interactions to only exist at unsampled extremes

Area-level vs. individual-level measurement

- geospatial exposures are proxies
- introduces ecological fallacy risk

Temporality/single time-point measurement

- only using baseline exposure can lead to time-varying misclassification

Data quality (EHR)

- coding heterogeneity
- missingness not at random, etc.

Difficulties for cancer GxE specifically

- Interaction effects require LARGE sample sizes
- Most cancers are rarer than traits like BMI, height, etc.
- Long latency between exposure and diagnosis leads to frequent limitations for single-timepoint exposure assessments



Replication (lessons from psychiatric epi)

- **Claim**: 5-HTTLPR short-allele carriers showed elevated risk of depression after stressful events, while long/long genotype was protective (total $n \sim 800$) (Caspi et al. 2003 *Science*)
 - *highly cited, motivated many candidate GxE studies for other traits*
- **Test**: Risch et al. did a meta-analysis of 14 studies in 2009 (*JAMA*) and found no genotype main effect and no gene-by-stress interaction (total $n \sim 14,000$)
- Small, underpowered candidate GxE studies can generate compelling-looking interactions that may not replicate at scale.

Opportunities for progress

- Many flagship hits so far have been in lung function, cardiovascular disease, and metabolic traits
- New methods to identify candidate loci without specifying scale and to model environmental exposures as composite measures
- Better exposure measurement can refine results

The future of GxE research

- Coordinated harmonization of environmental variables
- Pooling and consortia efforts to increase sample size and power for GxE detection

These advances can enable us to ask new and refined questions that we can then answer with better data!



Analysis and practicum

Common analytic approaches: 2x2x2 table (standard)

	E = 1 (smoker)		E = 0 (non-smoker)	
	G = 1	G = 0	G = 1	G = 0
Case	60	100	40	200
Control	40	160	40	160

- What is the GxE OR?
 - $OR(G|E=1) = (60 \times 160) / (100 \times 40) = 9,600 / 4,000 = \mathbf{2.4}$
 - $OR(G|E=0) = (40 \times 160) / (200 \times 40) = 6,400 / 8,000 = \mathbf{0.8}$
 - Interaction OR = $2.4 / 0.8 = \mathbf{3.0}$

Case-only designs

- 2 assumptions: disease is rare and G and E are independent
 - the OR comparing individuals exposed to G and E to individuals exposed to neither G nor E = the product of the OR of individuals exposed to only one or the other G or E
 - efficient if assumptions hold
-
- but, if G and E are *not independent* in non-cases (e.g. population stratification), the case-only estimate is biased

Common analytic approaches: 2x2x2 table (case-only design)

	E = 1 (smoker)		E = 0 (non-smoker)	
	G = 1	G = 0	G = 1	G = 0
Case	60	100	40	200
Control	40	160	40	160

- What is the GxE OR?
 - First, check independence in controls.
 - $OR(G-E \mid \text{controls}) = (40 \times 160) / (160 \times 40) = \mathbf{1.0 \text{ (good)}}$

Common analytic approaches: 2x2x2 table (case-only design)

	E = 1 (smoker)		E = 0 (non-smoker)	
	G = 1	G = 0	G = 1	G = 0
Case	60	100	40	200

- What is the GxE OR?
 - Next, restrict to cases only.
 - Case-only OR = $(60 \times 200) / (40 \times 100) = 12,000 / 4,000 = 3.0$

same OR in this case, but tighter confidence interval!

Sample size

- GxE studies require large sample sizes (~4x for main effects)
 - interaction effect estimates have much larger variance
- Low power can limit replicability (Caspi et al. example)
- Broadening sampled exposure range or using aggregate genetic measure can help

Testing additive vs. multiplicative interactions

- Multiplicative scale: test $H_0: \text{Beta}_{GE} = 0$ in the log-odds model
- Additive scale: test H_0 on the risk-difference scale, and report RERI
- Always report which scale you tested!

Thank you!

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